Data Science Research Project

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By

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Short-Term Solar PV Forecasting in NSW Using Machine Learning

Declaration

I, **Kingsley Ogadinma Onyekwele** hereby declare that the work presented in this thesis is the result of my own effort and has not been submitted, either in whole or in part, for any other degree or qualification. Where the work of others has been used, it has been duly acknowledged in the text and listed in the references. This report does not contain material previously published or written by another person, except where due reference is made. Any views expressed are those of the author and do not necessarily reflect those of the Sydney Polytechnic Institute, or any associated organisation.

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I also thank the faculty of the Master of Data Science program at the Sydney Polytechnic Institute for the grounding in machine learning, statistical practice, and research method that this project draws on, and my fellow students in the MDS650 cohort, whose questions during the progression reviews sharpened the research question and kept the project's difficulties in proportion. The progress presentations in particular benefited from colleagues who asked what the results would mean for an operator rather than only for a model.

The solar generation and market data used in this work were obtained from the Australian Energy Market Operator through the NEMOSIS dynamic data compiler, and the meteorological reanalysis from renewables.ninja. I acknowledge both services, and the underlying MERRA-2 project of NASA's Global Modeling and Assimilation Office, for making these datasets openly available. Open data infrastructure of this kind is what allows a reproducible regional benchmark to be assembled, re-executed, and extended by others, and this project depends on it end to end.

Finally, I thank my family for their patience and encouragement across this trimester and throughout the Master of Data Science course.

Abstract

This study evaluates whether scale-normalised five-minute intra-hour variability improves short-term forecasts of AEMO NSW1 market-cleared solar generation after ordinary hourly generation history is controlled. Five-minute AEMO observations were aggregated into complete hours and aligned with regional hourly MERRA-2 reanalysis. Exact duplicate timestamps were collapsed and conflicting duplicates were removed. XGBoost, LSTM and a physics-guided CNN-BiLSTM were compared with persistence, rolling climatology and SARIMA at 1, 3, 6 and 24-hour horizons using a chronological 2025 test set. Neural headline forecasts average five seeds. Robust findings require agreement across a 168-hour moving-block confidence interval, three corrected Diebold-Mariano bandwidths and a past-only output-scale sensitivity check.

All three learned models obtained robust intra-hour feature gains at one hour. XGBoost and the physics-guided model also improved robustly at three and six hours, while LSTM did not. No robust benefit occurred at 24 hours. Robust controlled gains ranged from 1.12 to 4.88 percent, substantially smaller than the 43.21 to 67.75 percent XGBoost gains from adding ordinary hourly history. On combined inputs, LSTM achieved the best one-hour RMSE of 155.25 MW; the physics-guided model was best at three, six and 24 hours with 300.28, 369.17 and 400.81 MW. The neural models were not robustly different at one to six hours, while the physics-guided model was robustly better at 24 hours. The weighted ensemble did not outperform the best neural model.

The evidence supports five-minute variability as a modest short-horizon supplement, not the main source of predictive skill. Conclusions are limited to one held-out year, retrospective weather inputs and utility-scale market-cleared generation. The final notebook completed from a fresh kernel without errors on Apple Silicon using PyTorch MPS.

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1 Introduction

1.1 Background and Research Context

New South Wales (NSW) provides a regional setting for studying short-term solar photovoltaic (PV) forecasting. Variation in solar output complicates the balance between generation and demand. Forecasts can support scheduling and dispatch by helping the operator anticipate that variation [1]. This project examines predictive accuracy; it does not estimate avoided curtailment, balancing costs or financial benefits.

The value of a forecast depends on its lead time and the decision it supports. The study evaluates one-hour, three-hour, six-hour and day-ahead horizons to cover intraday and next-day planning. These are experimental forecasting horizons, rather than direct simulations of AEMO scheduling or settlement procedures. Regional MW errors alone cannot establish participant-level costs or savings.

Relevant work includes hierarchical temporal convolutional networks for day-ahead regional forecasting in Western Australia [2], transformer models for irradiance and PV power forecasting [3], [4], characterisation of intra-hour solar radiation variability [5], and satellite-based prediction at four Australian locations [1]. These studies do not supply the matching NSW1 benchmark sought here. NSW exposure to East Coast Low events [6] provides regional context, although this project does not evaluate event-specific forecast errors.

1.2 Research Problem and Motivation

The problem motivating this project is the lack of an identified NSW benchmark in the reviewed literature that combines local meteorological reanalysis with grid data for short-term solar PV generation forecasting. Prior Australian work has forecast at regional or state level using weather data and historical generation, and utility-scale forecasts feed national balancing through AEMO's operational solar forecasting processes, but the retained review did not identify the same NSW1 target, source combination and evaluation design. The distinction is one of benchmark coverage rather than an absolute claim that no NSW-related forecasting work exists.

A second constraint is the meteorological source. MERRA-2 is retrospective reanalysis rather than an operational forecast [7]. The experiment uses weather information associated with the forecast origin and its history, and does not substitute archived numerical weather predictions issued in real time. Its value therefore lies in comparing input sets and configurations within a retrospective benchmark. Neither a fixed deployment-error penalty nor unchanged operational model rankings can be inferred from this design.

The study therefore provides a regional reference and tests which input information contributes to accuracy under a shared retrospective evaluation.

1.3 Research Questions and Objectives

The research question guiding this project is: to what extent do scale-normalised AEMO five-minute intra-hour variability features improve forecast skill after controlling for ordinary hourly generation history?

Three objectives structure the work. The first is to benchmark XGBoost, a unidirectional LSTM and a physics-guided CNN-BiLSTM against persistence, rolling climatology and seasonal ARIMA (SARIMA) over four horizons: 1 h, 3 h, 6 h and 24 h. The second is to estimate the incremental contribution of two scale-normalised intra-hour variables after holding ordinary hourly generation history constant. The third is to compare the learned forecasting procedures and describe whether a validation-weighted ensemble improves their point accuracy. Each objective uses the same 2025 evaluation year and horizon-specific common forecast windows.

The primary hypothesis is that adding the coefficient of variation and relative five-minute range to a weather-plus-hourly-history input set reduces RMSE. A controlled gain is treated as stable only when the 168-hour moving-block interval excludes zero, Diebold-Mariano tests remain significant after correction under all three dependence bandwidths, and an output-scale-relative sensitivity check agrees in direction. A broader XGBoost factorial comparison estimates the larger contribution of ordinary hourly history and the value of intra-hour features without that history.

The architecture comparison assesses forecasting procedures rather than isolated components: the physics-guided model changes the convolutional front end, recurrence direction and loss together. The controlled input comparison answers the research question, while the ensemble and weather-regime analyses remain secondary descriptions.

The dataset underpinning these objectives combines hourly renewables.ninja meteorological reanalysis [3, 8] with 5-minute AEMO NEMOSIS dispatch data [9] for September 2020 to 2025, engineered into four controlled information sets containing up to 16 variables, and split chronologically into training (2020 to 2023), validation (2024), and test (2025) with 46,570 merged observations and 46,558 after feature burn-in.

1.4 Significance and Expected Contribution

The contribution is an applied NSW benchmark combining meteorological reanalysis [3, 8] and AEMO generation history [9]. It compares established architectures and uses a controlled four-set design to distinguish ordinary hourly history from intra-hour variability. Regime-conditioned errors describe where measured gains occur, but remain exploratory. The novelty claimed is limited to the benchmark and comparison design identified through the reviewed literature, rather than a new forecasting architecture.

For AEMO and practitioners, the benchmark provides a retrospective comparison of input choices and established configurations. The target is market-cleared dispatch for NSW1 and excludes most rooftop production. Operational application would require archived issue-time

weather forecasts, latency checks and a missing-data policy. Extending the benchmark to distributed PV would require distributed observations.

The evaluation distinguishes the value of all generation-history predictors from the incremental contribution of their within-hour dispersion.

This distinction matters for both scientific interpretation and system design. If a model improves after all AEMO history is added, the change may simply reflect access to the latest generation level, which is an unusually strong predictor of near-term output. It does not show that the five-minute sampling frequency is responsible. Conversely, a controlled comparison between otherwise identical hourly-history inputs can estimate whether variability within the latest completed hour carries information that its mean and trailing summaries omit. The final design therefore treats information-source attribution as the principal contribution and model ranking as a related benchmark result.

The four forecast horizons represent distinct decisions rather than interchangeable repetitions. One and three hours emphasise near-term cloud and ramp persistence, six hours crosses a substantial part of the daylight cycle, and 24 hours tests day-ahead periodicity. Results are therefore interpreted by horizon, not collapsed into one overall score.

The study also separates retrospective validity from operational readiness. A chronologically held-out year provides a meaningful test of temporal generalisation, but reanalysis weather, archived market data and zero assumed delivery delay do not recreate a live forecast service. The report accordingly uses terms such as benchmark, association and conditional skill. It avoids claiming deployable accuracy, economic savings or causal effects of weather regimes. These boundaries are part of the research design rather than qualifications added only after observing the results.

1.5 Thesis Outline

Chapter 2 synthesises the literature and research gap. Chapter 3 describes the data, models and evaluation design. Chapter 4 presents the results, Chapter 5 interprets them and their limitations, and Chapter 6 states the contribution and future work.

1 Introduction	Research question, NSW context, objectives and scope.
2 Literature review	Data, algorithms, constraints and evaluation practice.
3 Methodology	Controlled features, models, alignment and robust decision rules.
4 Results	Baselines, learned models, controlled gains and stability.
5 Discussion	Interpretation, relation to literature and limitations.
6 Conclusion	Findings, contribution and future validation needs.

Figure 1.1: Thesis progression from the research question to conclusions.

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2 Literature Review

2.1 Introduction

This chapter reviews short-term solar photovoltaic generation forecasting with machine learning, focusing on data, model choice and evaluation relevant to New South Wales (NSW). The retained literature did not identify a benchmark matching this project's NSW1 market-cleared target and combination of reanalysis with completed-hour AEMO generation history.

The review draws on the project's July 2026 search notes, which record 15 Boolean arXiv queries and three Semantic Scholar queries, supplemented by citation tracking and Crossref searches. The notes describe a retained set of 26 studies and 12 key references, but do not provide a complete auditable screening ledger. The reported arXiv corpus total is not a screened-paper count. This is a focused narrative review; its gap statement is bounded by the retained material and does not establish the absence of all related NSW research.

Section 2.7 integrates the four themes into the research gap and the main input-history question and the supporting architecture and ensemble comparisons.

2.2 Data Characteristics and Preprocessing

Two distinctions structure the data literature. Pfenninger and Staffell [7] evaluate hourly PV simulations based on reanalysis and satellite data in Europe. Ramirez Camargo and Schmidt [10] compare ERA5-land with MERRA-2-based renewables.ninja output at 57 Chilean plants, finding only modest improvements under their modelling assumptions. These studies motivate historical data use without validating this NSW meteorological aggregation. By contrast, Nielsen et al. [11] use satellite sequences for short-term irradiance prediction. The second distinction is temporal: hourly meteorology [12] and five-minute AEMO dispatch [9] require explicit alignment. This project retains completed-hour dispersion summaries alongside hourly history.

Multi-source studies address different information needs. Rabault et al. [13] combine reanalysis and satellite products to improve GHI estimates against Norwegian observations; Perera et al. [2] combine generation and weather across 101 Western Australian PV locations; Prasad and Kay [1] convert satellite-derived irradiance forecasts into power at Queensland and Victorian sites. Wen et al. [14] combine historical, meteorological and forecast variables with cross-dimensional feature extraction. These are related uses of complementary inputs, not a matched test of AEMO history. Likewise, validation of long-term hourly output and prediction of moving clouds answer different questions; their findings do not demonstrate a contradiction about whether one spatial resolution is universally sufficient.

2.3 Algorithms and Modelling Approaches

The modelling literature spans statistical methods, gradient boosting, recurrent networks and newer transformer or constrained hybrids. The relevant question is whether added complexity improves a comparable forecasting task.

Architecture rankings depend on the task and comparison. Mukhoty et al. [8] report strong LSTM encoder-decoder performance against their forecasting alternatives, while Mollasalehi and Farhadi [15] review model prevalence, complexity and execution time rather than establishing a universal accuracy-to-speed winner. Feng et al. [16] report benefits from clustering and model blending, and Perera et al. [17] report a 3.81 percent average MASE reduction from forecast combinations against their best individual model. Abdullah and Mohammed [18] report GHI RMSE of 19.53 W/m2 versus 30.64 W/m2 for attention-based alternatives in Sudan. Their physical guidance includes engineered inputs, so this result does not validate the present non-negativity loss. Hoang [19] reports transformer R2 of 0.9696, whereas Scott et al. [4] find near-baseline performance with poorer sunny-day results. SolarTformer [20] and CT-PatchTST [21] investigate attention and input relationships, while Sudharshan et al. [22] combine SARIMA and BiLSTM for seasonal irradiance prediction. The differing targets and protocols prevent a common model ranking.

2.4 System and Architecture Considerations

Operational constraints receive less attention than accuracy in the reviewed work. Dao et al. [23] consider ensembles on a Raspberry Pi, while Schubnel et al. [3] use a more demanding multimodal model. Their settings illustrate why deployment cost and available inputs belong alongside accuracy in model selection.

The staged baseline, tree and recurrent comparison tests incremental gains within the project's resources. Transformers are excluded by scope, not evidence of inferiority. The complexity claim in Abdullah and Mohammed [18] motivates comparison, but one study of engineered physical inputs is not evidence that the present loss penalty causes an improvement.

2.5 Evaluation Metrics and Experimental Design

Point-error metrics such as RMSE support model comparison, but do not describe the full distribution of future generation. Voyant et al. [24] motivate evaluating more than one error characteristic. Probabilistic work is already present in the retained literature: SolarCrossFormer [3] produces probabilistic irradiance forecasts. The present study evaluates point forecasts and uncertainty in aggregate performance; its bootstrap intervals are not predictive intervals for individual future hours.

Evaluation choices limit comparisons across studies. Perera et al. [2] report a statistically significant improvement, so significance testing cannot be presented as absent from the re-

viewed literature. Pedro and Coimbra [25] explicitly include persistence and ARIMA, supporting the use of strong target-only baselines here. Scott et al. [4] identify sunny-day degradation, while Prasad and Kay [1] show that delivery latency affects live forecasts. These findings motivate baseline comparison, paired uncertainty assessment and explicit input timing in the present benchmark. Irradiance and generation also remain distinct targets: Prasad and Kay [1] model their conversion, whereas this project directly forecasts market-cleared MW. Cross-study errors cannot therefore be treated as comparable effect sizes.

2.6 Conceptual Framework

The conceptual framework separates input information from model configuration. Data-driven models learn relationships between weather, calendar variables, generation history and the target. Physical guidance can instead enter through engineered variables, as in Abdullah and Mohammed [18], or through a constraint in training. These mechanisms should not be conflated. Voyant et al. [26] find that a clear-sky-free forecasting approach can rival or outperform clear-sky-based alternatives, which cautions against assuming that additional physical preprocessing always improves accuracy. Here, the history ablation compares information sets, while the constrained CNN-BiLSTM comparison changes both architecture and loss.

On the data side, multi-source approaches combine meteorology and generation within a feature space [2], [1], [13], [14]. The present benchmark tests both the input contribution and architecture differences under a common evaluation.

2.7 Research Gap and Synthesis

The reviewed evidence supports testing data integration while treating architecture gains as task-dependent. Differences in targets, input availability and evaluation protocols prevent a universal ranking [2], [4], [15].

The gap is the lack of a matching comparison in the retained study set. Perera et al. [2] forecast distributed rooftop generation in Western Australia; the present target is regional market-cleared solar dispatch in NSW1. Several studies combine inputs [2], [1], [14], but do not supply the same comparison of meteorology-only, hourly generation history and additional AEMO dispersion features. This source-and-target distinction defines the applied contribution. It does not establish priority over every use of AEMO data or claim that geographic transfer alone constitutes a new method.

Transfer is limited by different targets and input availability. Much reviewed work predicts irradiance rather than market-cleared MW. Retrospective reanalysis differs from weather forecasts available at operational issue times, so neither a fixed deployment penalty nor unchanged model rankings is established. The present meteorology is not independently validated against local ground-station observations; published validation in other regions [7], [10] motivates the source choice but does not establish local accuracy.

The study joins reanalysis and AEMO data for NSW, compares completed-hour history inputs, and adds paired significance tests to point-error metrics. Distributed forecasting remains distinct: the rooftop settings studied by Perera et al. [17] are outside this market-cleared target. The benchmark therefore supports regional dispatch comparisons without claiming coverage of total NSW solar production.

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3 Research Methodology

This chapter describes the final, executable forecasting design. Hourly MERRA-2 reanalysis obtained through renewables.ninja [2, 3] was aligned with five-minute AEMO NSW1 solar cleared generation [9]. Three learned model families, three baselines and a descriptive weighted ensemble were evaluated at 1, 3, 6 and 24-hour horizons. The principal experiment compares weather_history with combined inputs so that the incremental value of five-minute intra-hour variability is measured after controlling for ordinary hourly generation history.

Acquire and align	Fixed-AEST AEMO clock; complete hourly bins; join regional MERRA-2 at UTC instants.
Resolve duplicates	Collapse identical values; drop conflicting timestamp groups and incomplete hours.
Engineer predictors	Weather, ordinary hourly history and scale-normalised intra-hour variability.
Train and validate	Three learned models; three baselines; four direct forecast horizons.
Evaluate evidence	Common windows, five neural seeds, paired tests and moving-block intervals.
Record execution	Fresh kernel, Apple Silicon MPS, package versions and fixed random seeds.

Figure 3.1: Research workflow implemented by the notebook.

3.1 Data Sources and Target

The target is SS_SOLAR_CLEAREDMW from AEMO's DISPATCHREGIONSUM archive, filtered to NSW1. It is market-cleared utility-scale solar generation, not total NSW solar production: most rooftop PV is absent, while curtailment and other dispatch effects remain in the target. Intervention labels were not available in the legacy export. Hourly meteorology comprises regional means over eight NSW locations and includes irradiance, temperature, wind and cloud variables. The renewables.ninja simulated electricity series, AEMO UIGF, semi-scheduled aggregates and all target-time generation fields are excluded from predictors.

The AEMO market clock is interpreted as fixed AEST (UTC+10), including legacy records labelled UTC, before both sources are joined on genuine UTC instants. Five-minute interval-ending observations are assigned to right-closed hourly bins labelled by their left edge. An hour is retained only when every required AEMO variable has twelve observations. Night-time zeros are valid data; incomplete hours are not imputed as zero.

Duplicate handling is deliberately conservative. Of 3,532 duplicated timestamp groups, 2,619 contain identical target values and are collapsed to a single unambiguous observation. The 913 groups with conflicting target values are dropped in full rather than averaged, because the archive does not identify which row represents normal dispatch. The conflicts affect 120

hourly bins, which subsequently fail the completeness rule. This policy removes all duplicated timestamps while avoiding a synthetic target that may never have occurred.

After cleaning and joining, 46,570 complete hourly observations remain. The chronological split uses Sydney calendar years: September 2020-2023 for training, 2024 for validation and 2025 for testing. After the rolling-feature burn-in, the splits contain 29,016, 8,782 and 8,760 rows. There are 118 excluded hours inside training, two inside validation and none inside test.

The validation year serves model selection and early stopping, while the test year is reserved for final comparison. The split also reflects changing installed utility-scale capacity: a random partition would mix later high-output observations into training and understate the challenge of forecasting a growing fleet. Excluded hours are not interpolated; the sequence constructor makes those gaps explicit, sacrificing sample size to preserve the meaning of elapsed forecast time.

3.2 Feature Engineering and Forecast Alignment

Four feature sets isolate different information sources. weather contains 11 predictors: six meteorological variables, a clearness index, and sine/cosine encodings of hour and day of year. weather_history adds the latest completed hourly generation value plus 24-observation trailing mean and standard deviation, giving 14 predictors. weather_intra adds only two scale-normalised five-minute summaries, the coefficient of variation and relative range, giving 13 predictors. combined contains all 16 predictors.

The intra-hour measures are computed from the twelve five-minute observations in the latest completed hour. Dividing dispersion and range by the corresponding hourly level reduces their dependence on the expanding NSW solar fleet. Denominators are protected near zero and the resulting values are bounded in the implementation. These predictors remain retrospective completed-hour measurements; they do not use any observation from the forecast target hour.

The clearness index divides GHI by an extraterrestrial horizontal irradiance approximation using a representative NSW coordinate. It is clipped to [0, 1.5] and set to zero when the denominator is at most 20 W/m2. This is a regional proxy, not a physical ceiling on output.

Each model receives a contiguous 24-hour input window and predicts one scalar target at one horizon. Targets are matched by exact elapsed timestamps, not row position. Windows crossing a missing hour, a split boundary or a daylight-saving discontinuity are rejected. This prevents a nominal 24-row sequence from spanning more than 24 elapsed hours. At the 24-hour horizon, 28,490 training windows remain.

Feature construction follows forecast-time availability. Weather and calendar fields describe the issue-time sequence; generation features end at the latest completed AEMO hour, and trailing statistics are shifted before joining to future targets. Split-specific windowing prevents a recurrent context from importing observations across a split boundary. Assertions

check monotonic unique timestamps, exact horizon displacement, expected feature counts, finite transformed values and identical targets across compared feature sets.

3.3 Models and Baselines

XGBoost, an LSTM and a physics-guided CNN-BiLSTM form the learned benchmark. A separate model is fitted for every horizon. XGBoost hyperparameters are selected with 30 Optuna trials on combined inputs for each horizon, using at most 400 trees and early stopping after 50 unimproved validation rounds. The selected settings and estimator count are then held fixed across all four feature sets at that horizon, making feature comparisons conditional on a common tuning choice. XGBoost is a single fitted model per block.

Direct horizon-specific fitting avoids requiring one model to compromise among targets with different dynamics. It also makes the feature effect interpretable at each decision horizon. The cost is a larger training matrix and no sharing of information among horizons. For XGBoost, the 24-hour sequences are flattened so lag position remains explicit to the tree learner. For recurrent models, the same information remains ordered as time steps.

The LSTM has one recurrent layer with 64 hidden units. The physics-guided model applies a 32-filter one-dimensional convolution with kernel size three, ReLU activation and max pooling, followed by a one-layer bidirectional LSTM with 64 hidden units and a 128-unit prediction head. Its loss is mean squared error plus weight 0.1 on two penalties: negative output and departure from zero during known night-time target periods. The comparison therefore evaluates a composite architecture-and-loss variant; it does not isolate the convolution, bidirectionality or either penalty.

Neural predictors and targets are standardised using training-window statistics only. Training uses batch size 512, learning rate 0.001, at most 100 epochs and patience 20, restoring the best validation checkpoint. Each neural block is trained at seeds 42, 123, 456, 789 and 1024. Headline predictions average the five seed forecasts before scoring; the report separately gives the mean and standard deviation of individual-seed RMSE. Only weather_history and combined are trained for the neural models because these are the controlled comparison needed by the research question.

Persistence repeats the latest completed-hour generation value. Rolling climatology uses only previously available observations for the same target local hour, with a 365-observation trailing window and an explicit availability check. SARIMA(1,1,1)(1,0,0,24) is fitted on the training series, extended through validation, and updated with observations available at each test issue time before forecasting. All baselines are scored on the same recurrent common window as the learned models.

The descriptive ensemble combines XGBoost, LSTM and physics-guided forecasts using inverse-validation-RMSE weights normalised to one. Weights range from about 0.30 to 0.35.

The ensemble is not included in confirmatory significance families because it is derived from the same component forecasts and is not central to the research question.

3.4 Evaluation and Statistical Decision Rules

RMSE in MW is the primary accuracy measure. The report also gives MAE, bias, R2, daylight RMSE, persistence skill and two scale-normalised measures. Relative RMSE divides by a past-only output scale available at the issue time, while peak-normalised RMSE divides by an observed peak reference. The share of predictions clipped at zero is reported because non-negativity post-processing can hide otherwise negative forecasts.

Controlled feature gain is 100 x (RMSE_weather_history - RMSE_combined) / RMSE_weather_history; a positive value favours intra-hour predictors. Moving-block bootstrap intervals use 2,000 resamples and block lengths of 24, 48 and 168 hours. The 168-hour interval is the conservative headline because it preserves a full week of serial structure.

Paired Diebold-Mariano tests [27] use squared-error differences and Newey-West estimates at bandwidths h-1, 24 and 168 hours. Bonferroni correction is applied separately within each predeclared family and bandwidth: 12 controlled feature comparisons, 36 learned-model-versus-baseline comparisons and 12 pairwise learned-model comparisons. A finding is called robust only when the 168-hour bootstrap interval excludes zero, all three corrected Diebold-Mariano tests agree, and the past-only output-scale calculation gives the same direction. This rule avoids selecting a conclusion from one favourable dependence assumption. Effect size remains distinct from significance.

The three inferential families answer different questions. The controlled family evaluates the added intra-hour variables within each model and horizon. The baseline family asks whether each learned procedure improves on persistence, rolling climatology and SARIMA. The learned-model family compares architectures on combined inputs. Keeping the corrections separate avoids treating exploratory ensemble and regime plots as confirmatory hypotheses while still controlling multiplicity within each substantive question.

3.5 Reproducibility and Ethics

The final notebook was executed from a fresh kernel with code-cell execution counts 1-30 and no recorded errors or standard-error output. It ran from 10:27:05 to 10:46:36 UTC on 13 September 2026 on Apple Silicon (arm64) using PyTorch MPS. The saved environment was conda spunky, Python 3.12.12, PyTorch 2.5.1, XGBoost 3.0.1, pandas 2.3.3 and NumPy 2.1.3. The notebook records these values, split boundaries, seeds, device choice and data assertions alongside its outputs.

The project uses observational agency data without human subjects or personal identifiers. Its reproducibility is computational rather than exact across all hardware: floating-point order,

accelerator kernels and package versions can produce small differences. Data-source acknowledgement does not itself establish redistribution rights [2, 3, 4, 5].

<!– Citations use the report’s global reference numbering. The printed reference list is generated from report/latex/main.bbl. Chapter-local reference lists were removed because they used a superseded numbering and contradicted the printed list. –>

4 Analysis, Experiments and Results

All results in this chapter come from the clean Apple Silicon execution described in Section 3.5. The common 2025 test window is used throughout. Neural headline metrics score the average of five seed forecasts; individual-seed dispersion is reported separately.

4.1 Data Characteristics

The cleaned series contains 46,570 complete hours. Generation is zero-inflated, strongly diurnal and seasonally structured. Capacity growth and market constraints make the target non-stationary, so these plots describe market-cleared generation rather than an invariant physical production distribution.

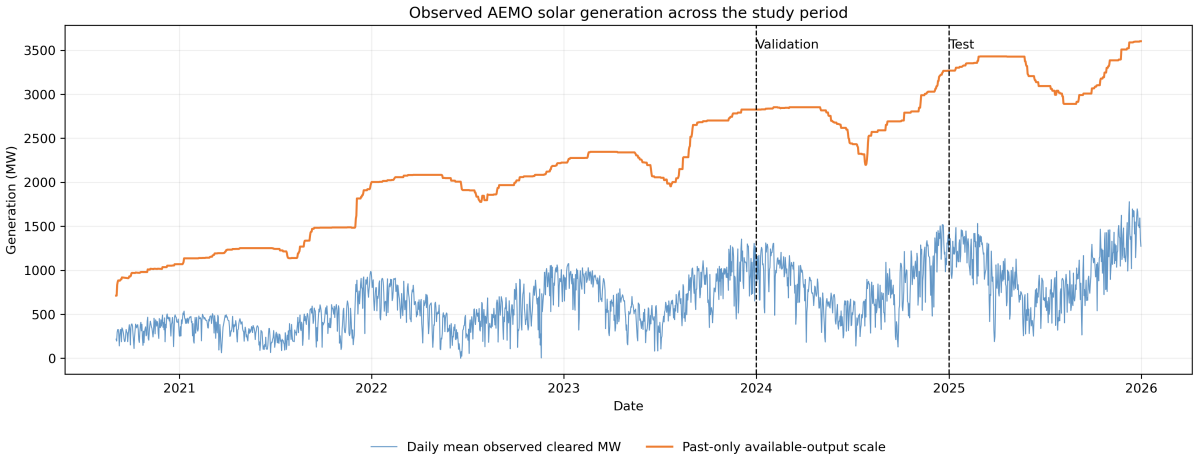


Figure 4.1: Hourly NSW1 market-cleared solar generation across the observation period.

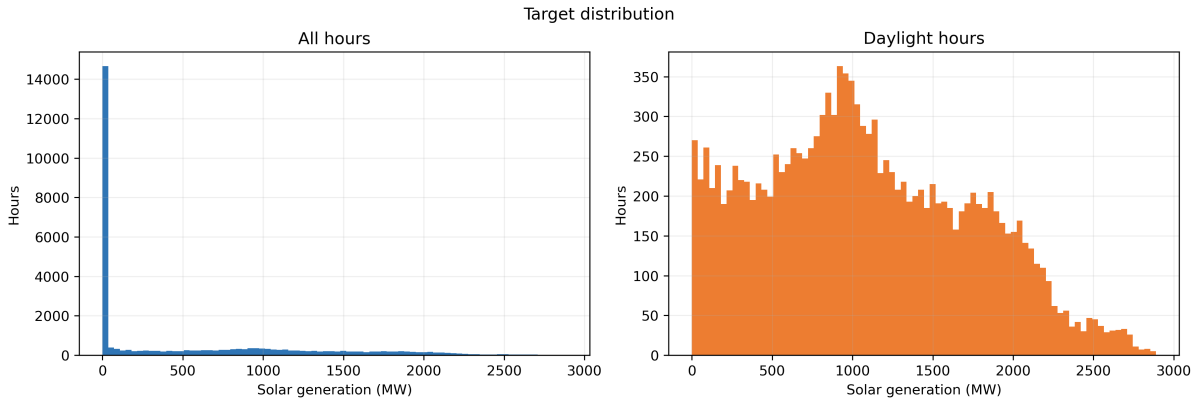


Figure 4.2: Distribution of hourly NSW1 market-cleared solar generation, including valid zero output.

Mean output follows the solar day, while the width and level of the seasonal profiles vary across the year. The split diagnostic shows that 2025 is not exchangeable with earlier years, supporting chronological rather than random evaluation.

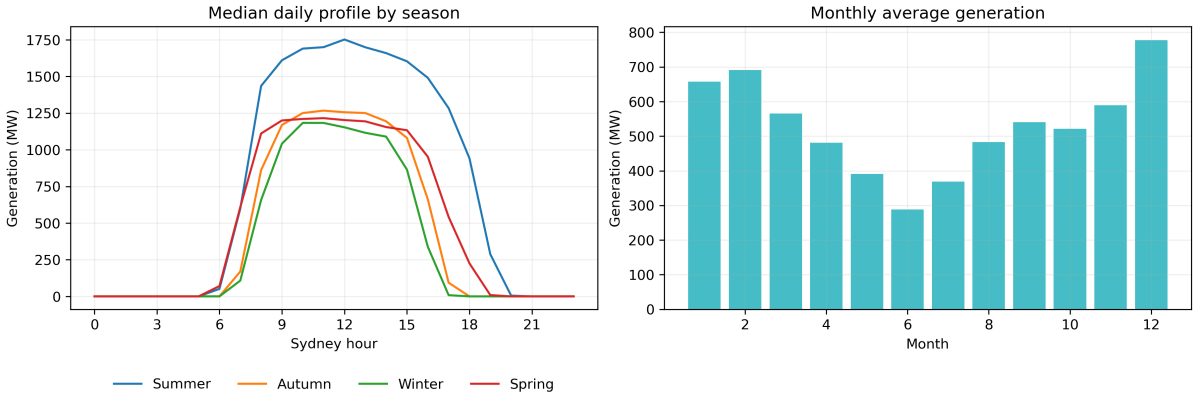


Figure 4.3: Hourly and seasonal structure in NSW1 market-cleared solar generation.

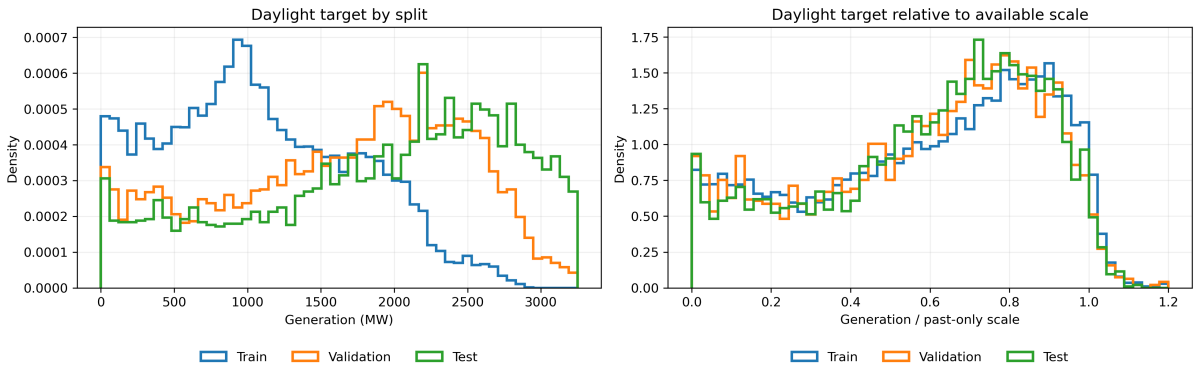


Figure 4.4: Target distribution across chronological training, validation and test splits.

Measured output also trends upward across the period, which matters for comparability between splits.

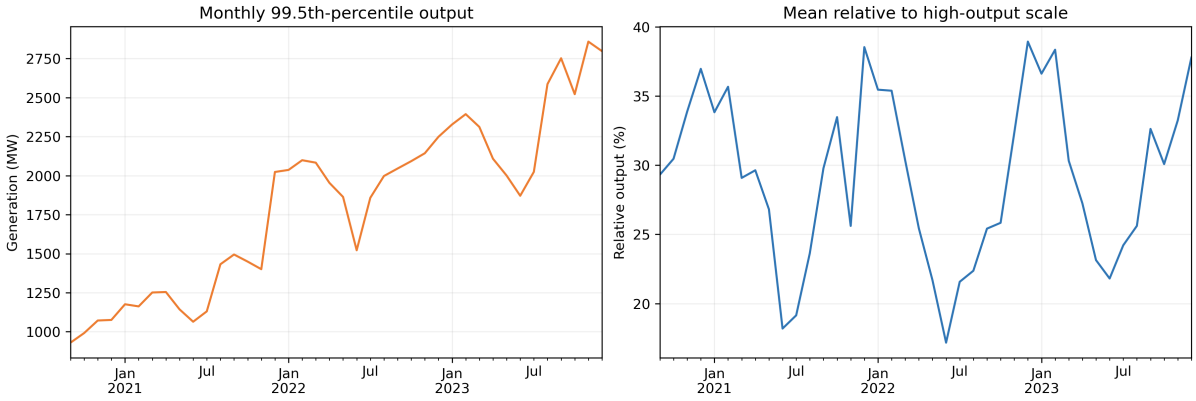


Figure 4.5: Monthly 99.5th-percentile observed output and mean generation relative to that scale. The level trend documents fleet growth that the past-only output scale is designed to absorb.

The five-minute source retains fluctuations hidden by hourly averaging. Scale-normalised coefficient of variation and relative range are the two candidate incremental predictors. Their relationship with the next-hour ramp motivates the experiment but is not itself causal evidence.

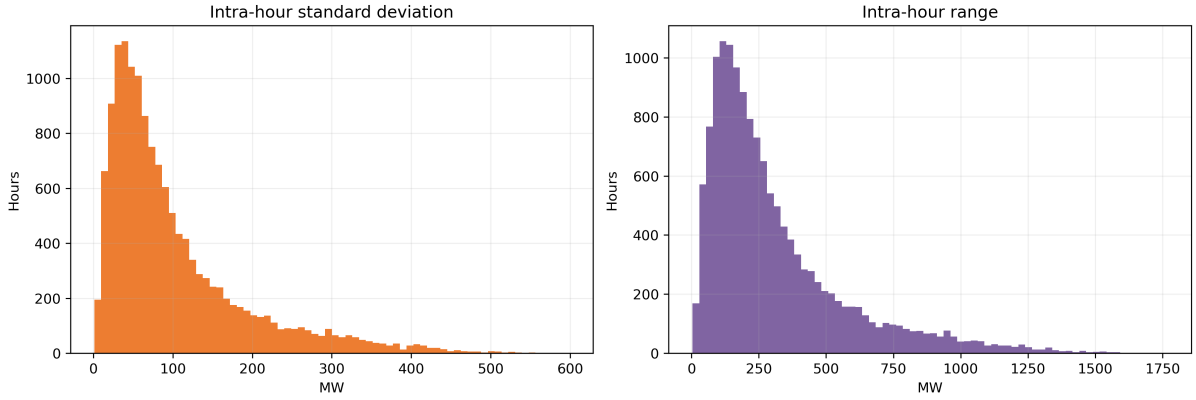


Figure 4.6: Within-hour variability retained from five-minute AEMO observations.

The exploratory sequence supports three decisions used later. First, zero output is retained because night-time behaviour is part of both the forecasting problem and the physics-guided loss. Second, the visible split drift justifies past-only scaling instead of normalisation by a full-sample maximum. Third, the monotone increase in next-hour change across variability bins makes an incremental short-horizon signal plausible. None of these observations selects a winning model; that decision remains based on the held-out paired comparisons.

4.2 Baseline Performance

Persistence is competitive at 24 hours because of daily periodicity, but weak at 3 and 6 hours across the daytime rise and fall. SARIMA is the strongest baseline at 1-6 hours, while rolling climatology is comparatively stable across horizons.

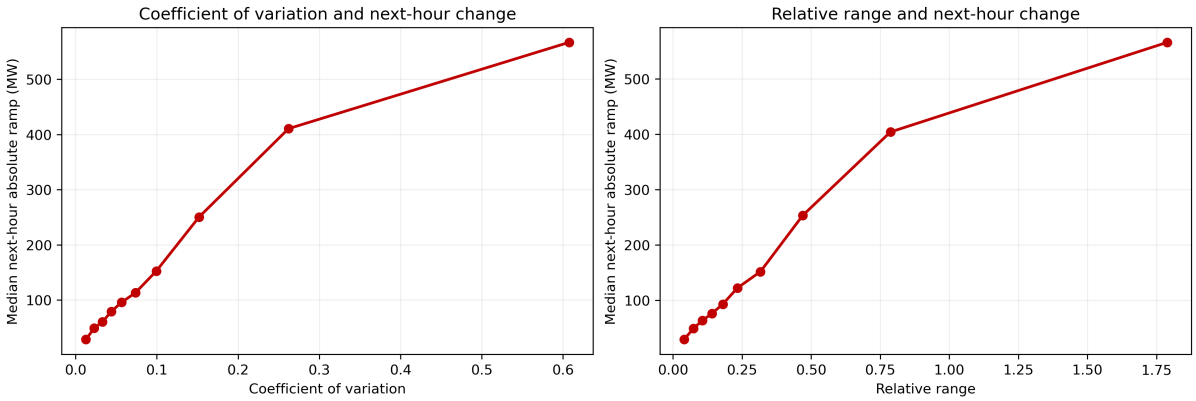


Figure 4.7: Association between scale-normalised intra-hour variability and the next-hour ramp.

The magnitude of the contrast is substantial. Persistence RMSE rises from 469.06 MW at one hour to 1,631.92 MW at six hours, then falls to 423.60 MW at 24 hours. This U-shaped pattern reflects the mismatch between copying the latest output across the intraday solar cycle and copying it to approximately the same time on the next day. Rolling climatology stays close to 571 MW because its same-hour historical average does not change its basic information with horizon. SARIMA achieves 221.69 MW at one hour and 433.32 MW at three hours, demonstrating that a target-only time-series model can be a demanding comparator even without weather fields.

Table 4.1: Baseline performance on the common 2025 test window. RMSE and MAE are MW.

Model	h	RMSE	MAE	R2	Persistence skill (%)
Persistence	1	469.06	242.10	0.84	0.00
Persistence	3	1092.77	658.82	0.13	0.00
Persistence	6	1631.92	1210.27	-0.94	0.00
Persistence	24	423.60	219.52	0.87	0.00
Rolling climatology	1	570.93	348.28	0.76	-21.72
Rolling climatology	3	570.99	348.36	0.76	47.75
Rolling climatology	6	571.09	348.47	0.76	65.01
Rolling climatology	24	570.37	347.76	0.76	-34.65
SARIMA	1	221.69	118.13	0.96	52.74
SARIMA	3	433.32	245.14	0.86	60.35
SARIMA	6	542.53	343.98	0.79	66.76

Table 4.1 continued

Model	h	RMSE	MAE	R2	Persistence skill (%)
SARIMA	24	634.49	330.15	0.71	-49.78

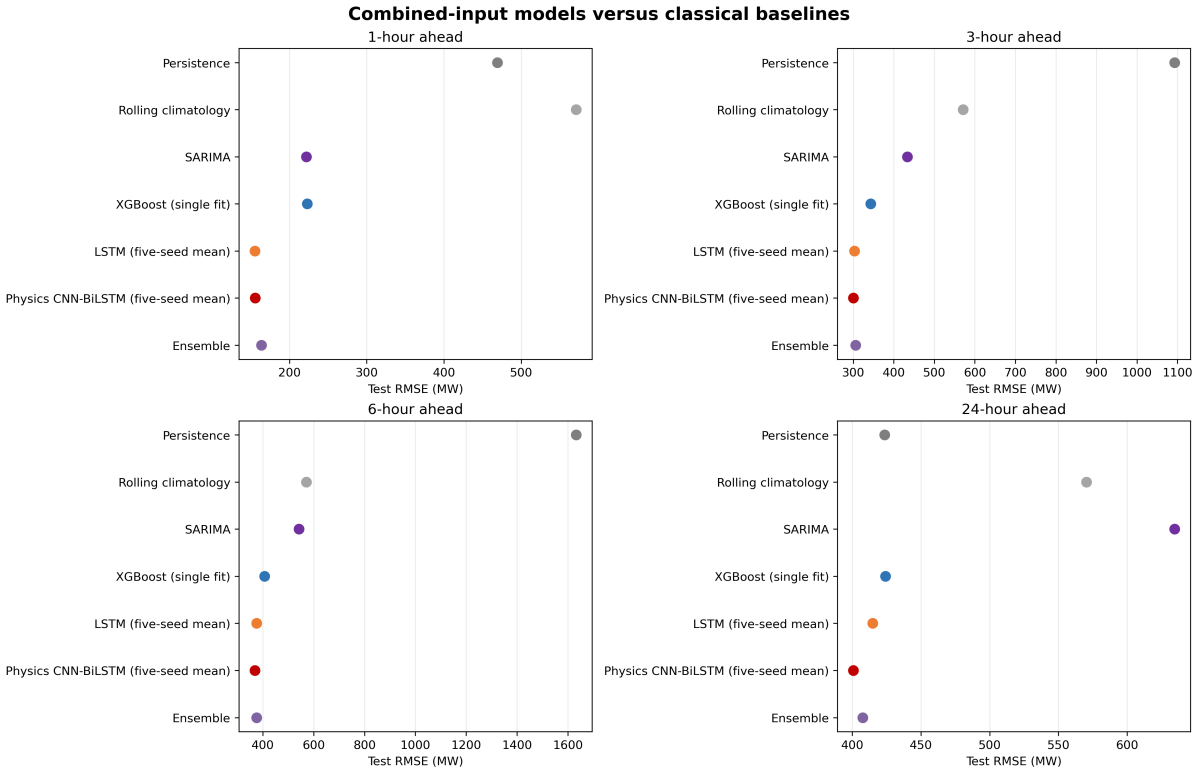


Figure 4.8: RMSE of persistence, rolling climatology and SARIMA across forecast horizons.

Baseline skill relative to persistence must therefore be read with the horizon. Rolling climatology is 21.72 percent worse than persistence at one hour but 65.01 percent better at six hours. SARIMA improves on persistence by 52.74-66.76 percent at one to six hours, yet is 49.78 percent worse at 24 hours. A single baseline or an average across horizons would conceal these reversals.

4.3 Learned-Model Performance

On combined inputs, LSTM gives the lowest 1-hour RMSE, while the physics-guided CNN-BiLSTM is best at 3, 6 and 24 hours. The margins between the two neural models are small at 1-6 hours. XGBoost remains substantially better than persistence at 1-6 hours but not at 24 hours.

Absolute error grows with horizon for every learned model, but reference-relative skill does not change monotonically. XGBoost RMSE increases from 222.90 to 406.76 MW between one

and six hours, while its persistence skill rises from 52.48 to 75.07 percent because persistence deteriorates faster. The corresponding neural skill peaks near 77 percent at six hours. At 24 hours, all learned RMSE values remain near 400 MW and persistence skill falls to between -0.11 and 5.38 percent. The forecast problem is consequently not ordered simply from easiest to hardest by lead time.

Table 4.2: Combined-input learned-model and ensemble metrics. RMSE, MAE, bias and daylight RMSE are MW.

Model	h	RMSE	Relative RMSE (%)	MAE	Bias	R2	Daylight RMSE	Skill (%)
XGBoost	1	222.90	6.65	116.74	-65.26	0.96	318.34	52.48
LSTM	1	155.25	4.70	83.68	-9.05	0.98	220.43	66.90
Physics-guided	1	155.53	4.71	86.09	-18.53	0.98	221.46	66.84
Ensemble	1	163.73	4.93	89.84	-29.47	0.98	233.41	65.09
XGBoost	3	343.11	10.38	189.72	-111.99	0.91	490.08	68.60
LSTM	3	302.65	9.16	165.90	-52.09	0.93	430.89	72.30
Physics-guided	3	300.28	9.12	170.12	-50.85	0.93	426.52	72.52
Ensemble	3	305.72	9.26	170.03	-70.90	0.93	435.86	72.02
XGBoost	6	406.76	12.35	230.36	-143.32	0.88	580.77	75.07
LSTM	6	375.28	11.37	216.11	-90.88	0.90	533.77	77.00
Physics-guided	6	369.17	11.20	217.28	-78.09	0.90	523.53	77.38
Ensemble	6	375.28	11.38	215.94	-103.58	0.90	534.65	77.00
XGBoost	24	424.07	12.88	239.95	-121.17	0.87	605.83	-0.11
LSTM	24	414.86	12.59	239.90	-121.68	0.87	591.80	2.06
Physics-guided	24	400.81	12.19	231.78	-95.44	0.88	571.66	5.38
Ensemble	24	407.67	12.38	234.17	-112.58	0.88	582.17	3.76

The additional metrics give a consistent but more nuanced view. Combined-input relative RMSE is 4.70-4.71 percent for the neural models at one hour and approximately 12 percent at 24 hours. Peak-normalised RMSE follows the same ordering. Bias is negative for every reported configuration. XGBoost carries the largest negative bias at 1, 3 and 6 hours (-65.26, -111.99 and -143.32 MW), while at 24 hours all three learned models are strongly negative, from -95.44 MW for the physics-guided model to -121.68 MW for LSTM, and the ranking between them is

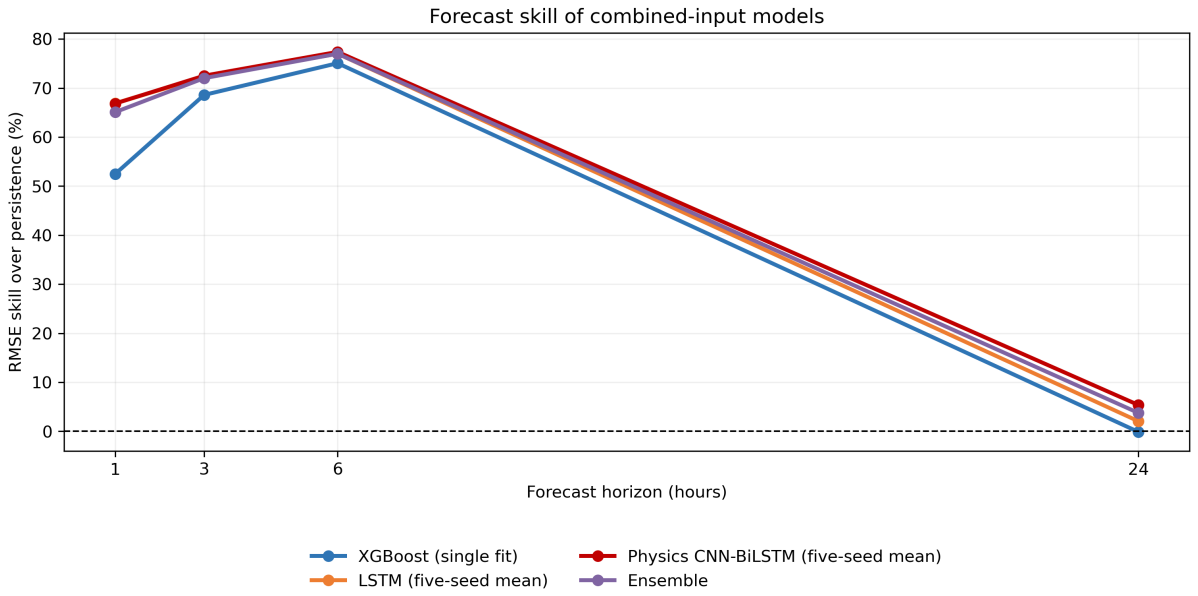


Figure 4.9: Persistence skill of learned models and the descriptive ensemble on combined inputs.

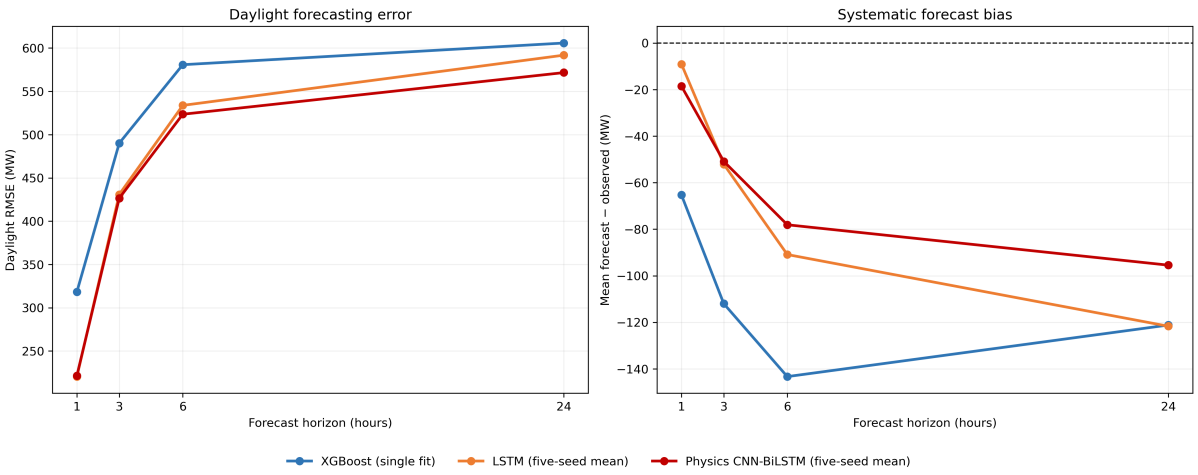


Figure 4.10: Daylight error and signed bias by horizon for the three learned models. Bias grows steadily with horizon, which the aggregate RMSE alone does not show.

no longer meaningful. Daylight RMSE is higher than all-hour RMSE because retained night-time zeros are comparatively easy. These differences explain why one scalar score should not be interpreted as uniform performance across the operating day.

Zero clipping also forms part of the measurement chain. At one hour, 16.71 percent of LSTM and physics-guided outputs are clipped, compared with 26.28 percent for XGBoost. Many clipped observations occur at night, where zero is physically appropriate, but clipping can also mask the magnitude of an unconstrained negative prediction. The report therefore treats the physics-guided loss and the final non-negativity operation as distinct: the former influences training, while the latter defines the reported forecast output.

The broad ranking must be qualified by paired inference. LSTM robustly outperforms XGBoost at 1, 3 and 6 hours, but not at 24 hours. The physics-guided model robustly outperforms XGBoost at all four horizons. Between the neural models, there is no robust difference at 1, 3 or 6 hours; the physics-guided model is robustly better only at 24 hours. Thus the point estimate at 1 hour favours LSTM, but it does not establish architectural superiority.

Effect sizes clarify this conclusion. Relative to XGBoost, LSTM gains 30.35 percent at one hour, then 11.79 and 7.74 percent at three and six hours. The 2.17 percent day-ahead gain is not robust. Physics-guided gains 30.23, 12.48, 9.24 and 5.49 percent over XGBoost, all robust under the report's rule. In contrast, neural-to-neural gains range from -0.18 to 3.39 percent. Only the 3.39 percent physics-guided advantage at 24 hours survives the full dependence and multiplicity checks.

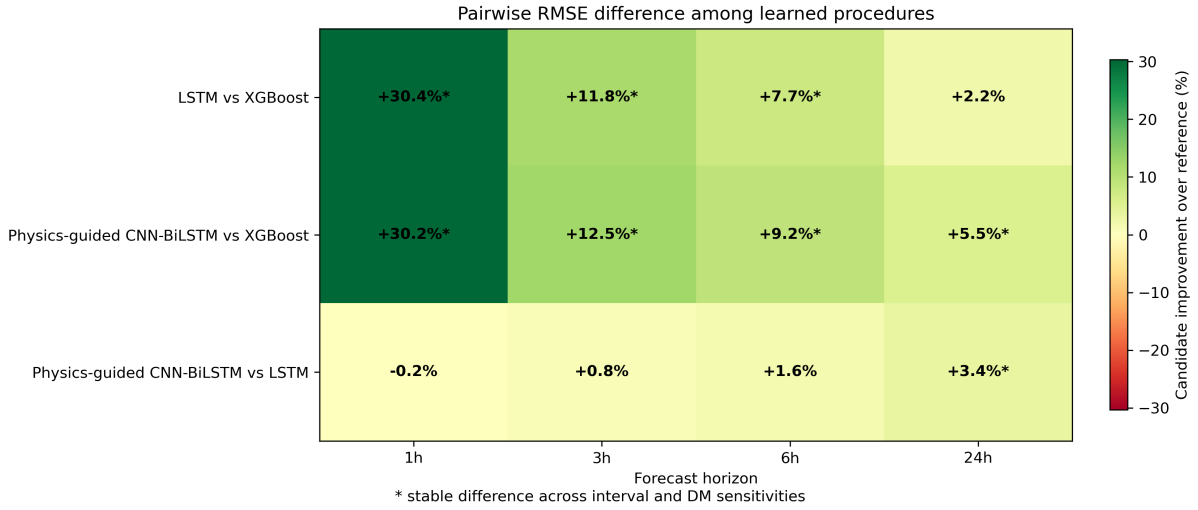


Figure 4.11: Dependence-robust pairwise conclusions among learned models.

Table 4.3: Pairwise learned-model RMSE gain of the second model over the first and robust conclusion.

Comparison	h	Gain (%)	Robust conclusion
XGBoost vs LSTM	1	30.35	LSTM better

Table 4.3 continued

Comparison	h	Gain (%)	Robust conclusion
XGBoost vs LSTM	3	11.79	LSTM better
XGBoost vs LSTM	6	7.74	LSTM better
XGBoost vs LSTM	24	2.17	No robust difference
XGBoost vs Physics-guided	1	30.23	Physics-guided better
XGBoost vs Physics-guided	3	12.48	Physics-guided better
XGBoost vs Physics-guided	6	9.24	Physics-guided better
XGBoost vs Physics-guided	24	5.49	Physics-guided better
LSTM vs Physics-guided	1	-0.18	No robust difference
LSTM vs Physics-guided	3	0.78	No robust difference
LSTM vs Physics-guided	6	1.63	No robust difference
LSTM vs Physics-guided	24	3.39	Physics-guided better

The falling neural advantage has a training-side explanation that the convergence records make visible. The epoch at which validation error is minimised falls sharply as the horizon lengthens.

Table 4.4: Mean best-validation epoch across the five seeds, by model, feature set and horizon. Lower values mean the fit stopped improving sooner.

Model	Inputs	1h	3h	6h	24h
LSTM	combined	61.2	31.0	18.4	8.8
LSTM	weather history	51.2	39.2	20.2	11.6
Physics-guided CNN-BiLSTM	combined	62.0	23.4	21.6	12.4
Physics-guided CNN-BiLSTM	weather history	60.0	26.2	22.4	12.0

The one-hour fits improve for roughly sixty epochs, whereas the day-ahead fits stop improving within about nine to twelve epochs and then overfit until patience expires. The recurrent models therefore extract far less structure from the same 24-hour input window at day-ahead range, which is consistent with the narrowing gap to XGBoost and with the negative controlled

gain at that horizon. The result concerns optimization behaviour under this fixed schedule rather than the information content of the inputs.

4.4 Controlled Contribution of Intra-Hour Features

The controlled comparison answers the research question directly. XGBoost shows robust positive gains at 1, 3 and 6 hours and no robust change at 24 hours. LSTM shows a robust gain only at 1 hour. The physics-guided model shows robust gains at 1, 3 and 6 hours; at 24 hours the combined inputs are worse, and the statistical result is not stable across the longest dependence bandwidth.

The confidence intervals illustrate why a deterministic ranking is insufficient. XGBoost's one-hour gain is 4.88 percent with a weekly-block interval from 3.79 to 6.09 percent, whereas its 24-hour interval spans -0.95 to 0.51 percent. LSTM's six-hour interval barely excludes zero, from 0.01 to 1.83 percent, yet its corrected Diebold-Mariano result is unstable when serial dependence is lengthened. Under the predeclared intersection rule, this remains inconclusive. Physics-guided at 24 hours shows a negative interval, but the longest-bandwidth corrected test does not remain significant; the safe conclusion is that no robust benefit from adding the features is established.

Table 4.5: Controlled intra-hour feature contribution relative to weather_history. Positive gain favours combined; confidence intervals use 168-hour blocks.

Model	h	RMSE history	RMSE combined	Gain (%)	95% CI (%)	Output-scale gain (%)	Finding
XGBoost	1	234.33	222.90	4.88	3.79 to 6.09	4.97	Robust improvement
XGBoost	3	346.98	343.11	1.12	0.51 to 1.70	1.11	Robust improvement
XGBoost	6	415.46	406.76	2.09	1.20 to 2.87	2.09	Robust improvement
XGBoost	24	423.00	424.07	-0.25	-0.95 to 0.51	-0.24	Inconclusive
LSTM	1	159.48	155.25	2.65	1.45 to 4.00	2.83	Robust improvement
LSTM	3	305.26	302.65	0.85	-0.22 to 1.99	0.99	Inconclusive
LSTM	6	378.67	375.28	0.89	0.01 to 1.83	0.92	Inconclusive

Table 4.5 continued

Model	h	RMSE history	RMSE combined	Gain (%)	95% CI (%)	Output-scale gain (%)	Finding
LSTM	24	411.98	414.86	-0.70	-1.32 to 0.06	-0.60	Inconclusive
Physics-guided	1	159.55	155.53	2.52	1.31 to 3.78	2.68	Robust improvement
Physics-guided	3	305.24	300.28	1.63	0.57 to 2.52	1.69	Robust improvement
Physics-guided	6	379.18	369.17	2.64	1.64 to 3.55	2.60	Robust improvement
Physics-guided	24	391.49	400.81	-2.38	-4.14 to -0.41	-2.23	Inconclusive across bandwidths

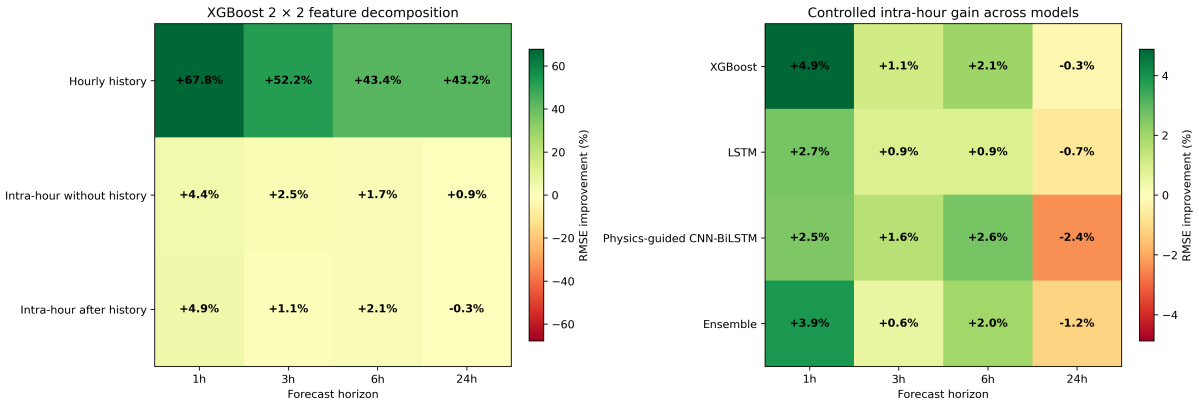


Figure 4.12: RMSE under weather, hourly-history, intra-hour and combined feature sets.

The feature-only XGBoost comparison reinforces the control logic. Intra-hour features added to weather alone gain 4.41, 2.49, 1.75 and 0.95 percent, whereas adding ordinary generation history gains 67.75, 52.22, 43.37 and 43.21 percent. Most of the broad benefit from AEMO history therefore comes from the latest level and trailing hourly summaries, not five-minute dispersion.

The two intra-hour estimates answer different questions. Measured against weather alone, the dispersion columns contribute between 0.95 and 4.41 percent. Measured against hourly history in Table 4.5, they contribute a similar 4.88 percent at one hour but only 1.12 percent at three hours. The closeness of the two figures at one hour is the informative part: the dispersion columns are close to redundant with the hourly history they accompany, because a completed hour's mean already reflects much of the variation inside it. That redundancy is why the controlled comparison, not the larger uncontrolled gain, is the evidence for the research question.

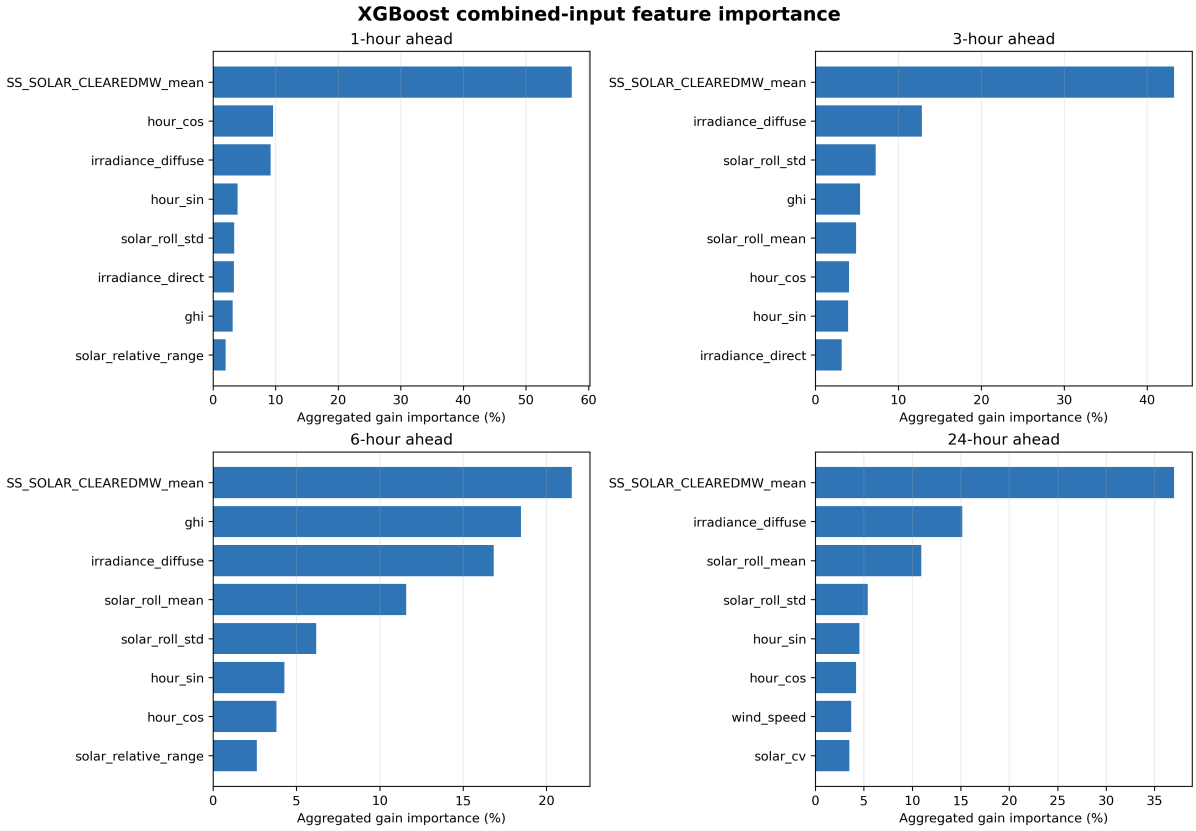


Figure 4.13: Combined-input XGBoost gain importance aggregated across the 24 lag positions. A descriptive diagnostic of model use, not a causal attribution or a significance test.

It also explains how the same two columns can appear substantial in one framing and marginal in another without either estimate being wrong.

The four-set design also shows interaction. Intra-hour features help XGBoost slightly when no ordinary history is present at all four horizons, but once hourly history is included the 24-hour change becomes negative. The short-horizon controlled gains are not merely a proxy for feature count: two additional variables can improve one model-horizon block and leave another unchanged. Because settings are held constant within horizon, these differences reflect the fitted use of the information under a shared tuning choice rather than a search favouring one feature set.

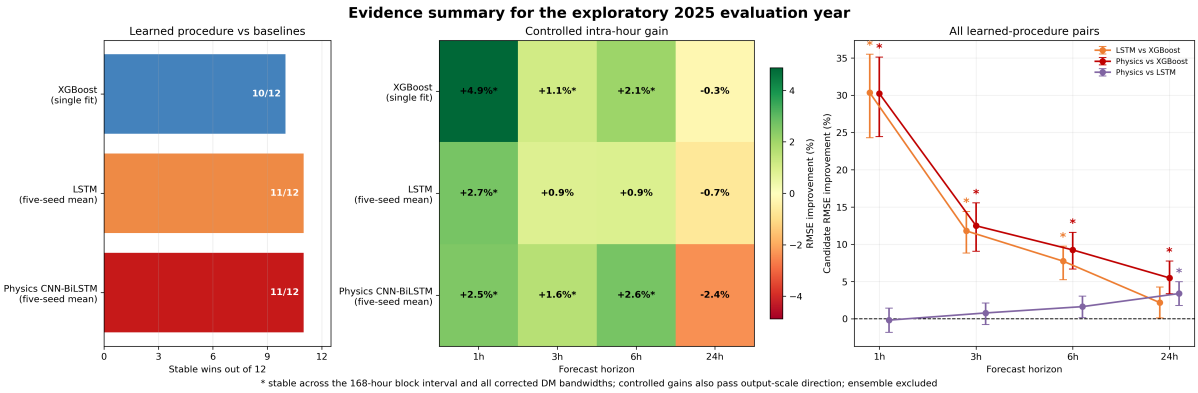


Figure 4.14: Summary of controlled intra-hour feature evidence across models and horizons.

4.5 Baseline Robustness, Weather Regimes and Stability

Across the 12 model-baseline comparisons for each learned family, XGBoost has 10 robust wins and each neural model has 11. All learned models robustly outperform persistence at 1-6 hours, but none does so robustly at 24 hours. All three robustly outperform rolling climatology at every horizon. XGBoost does not robustly beat SARIMA at 1 hour; both neural models do.

The count of robust wins is a summary, not a league table. The 36 baseline tests include comparisons with references of very different strength, and all horizons contribute equally to the count. The individual result remains the unit of interpretation. In particular, the absence of a robust 24-hour persistence win is more operationally relevant than another large six-hour win over a baseline known to fail across the intraday ramp.

Weather-regime analysis shows that the controlled feature effect is not uniform. Table 4.6 reports the gain for each model, horizon and daylight regime, where regimes are assigned from target-time clearness index and are never supplied to the models. These are secondary subgroup estimates: each regime covers roughly 1,700 to 2,000 hours, no multiplicity correction is applied within regimes, and repeated inspection of the same test year makes them less secure than the predeclared overall comparison.

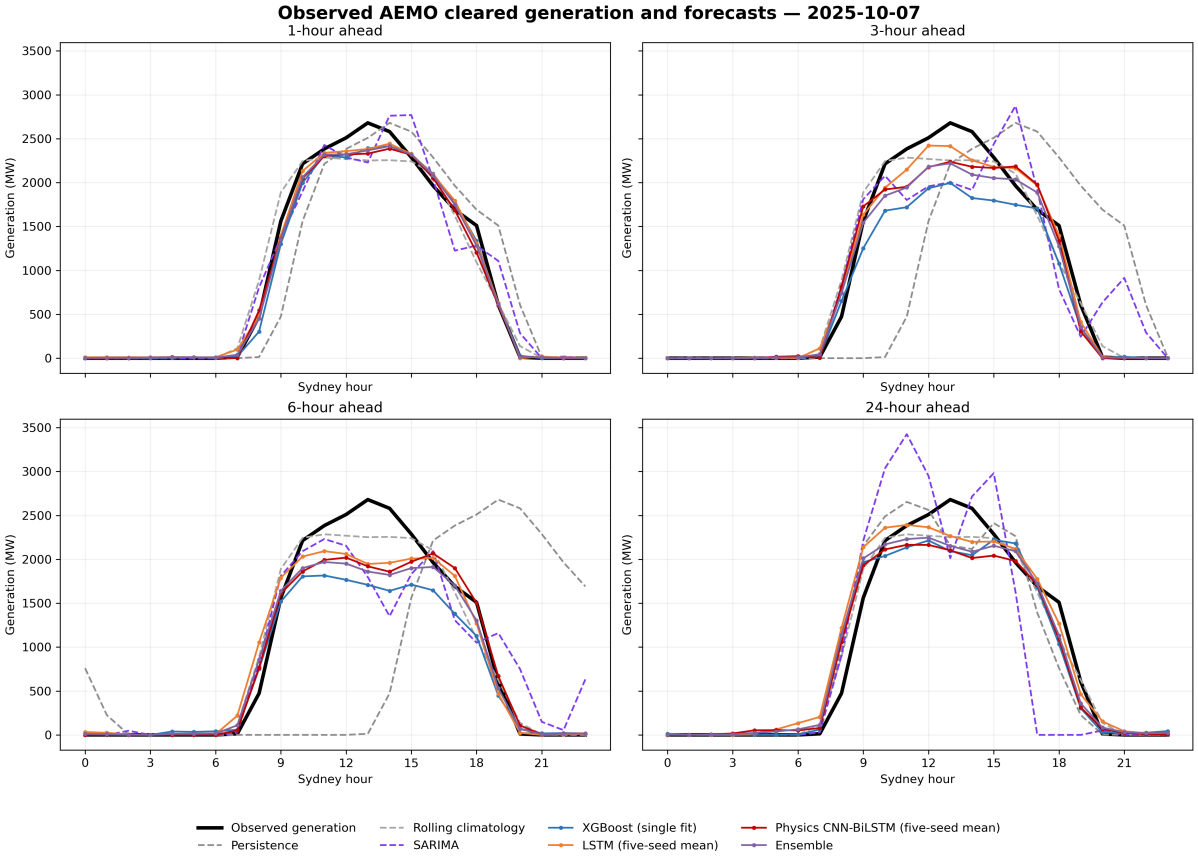


Figure 4.15: Observed and forecast trajectories for all models by horizon.

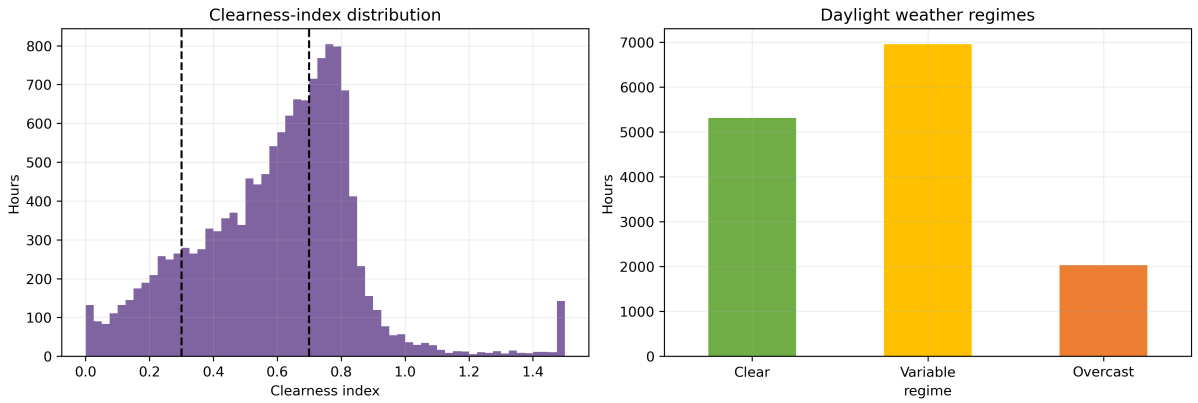


Figure 4.16: Clearness-index distribution in daylight hours and the resulting overcast, variable and clear regime counts. The thresholds define the strata used in the regime analysis and are evaluation-only.

Table 4.6: Controlled intra-hour RMSE gain (%) by daylight weather regime, from the predeclared weather_history versus combined comparison.

Horizon	Overcast	Variable	Clear	Range	Negative cells
1h	+2.46	+1.84	+5.58	-0.39 to +6.33	1 of 12
3h	+1.39	+0.64	+1.28	-2.22 to +3.53	2 of 12
6h	+0.19	+1.74	+2.40	-1.67 to +3.10	2 of 12
24h	+0.51	-0.54	-2.02	-3.94 to +3.85	9 of 12

Each cell is the mean across the four reported configurations (XGBoost, LSTM, physics-guided CNN-BiLSTM and the descriptive ensemble), so the table summarises direction rather than any single model.

The pattern contradicts the direction anticipated in the predeclared regime hypothesis. That hypothesis expected the largest gain under variable cloud, on the reasoning that coarse hourly reanalysis would miss rapid change and the dispersion features would then add the most. The measured ordering places clear conditions highest at 1 hour and again at 6 hours, and variable conditions lowest at 1 hour, where the overall controlled effect is strongest and robust for all three learned models. Variable conditions give the largest mean gain at no horizon and contain the only negative cell among the 1-hour results, from the physics-guided model at -0.39 percent. Of the five negative variable-regime cells, three fall at 24 hours, where the overall effect has already reversed.

A second reading fits the data better. Under broken cloud the hourly mean already reflects the cloud-driven variation within it, so a dispersion summary adds little beyond the level it is computed from. Under clear skies the hourly level is largely determined by solar geometry, so residual dispersion continues to carry information about departure from that expectation. On this reading the feature is most useful where the hourly history is least informative on its own, which is the opposite of what was predicted. The present design cannot separate that account from alternatives: clearness-index thresholds are arbitrary, eight-site averaging smooths the spatial variation that drives real cloud transitions, and regime counts are unequal.

Individual-seed RMSE confirms meaningful training variation, especially for longer-horizon neural fits. For combined inputs, mean individual-run RMSE is 160.78 +/- 3.25 MW for LSTM and 160.71 +/- 2.46 MW for the physics-guided model at 1 hour. At 24 hours it is 417.80 +/- 2.88 and 406.15 +/- 16.78 MW respectively. Forecast averaging yields the lower headline values in Table 4.2 and should not be confused with the mean of seed-level RMSE.

The physics-guided 24-hour standard deviation is notably larger than its one-hour value, indicating greater optimisation sensitivity despite its lower average forecast error. LSTM with weather_history also shows a 15.05 MW seed standard deviation at 24 hours. These observations support retaining seed ensembles for neural point forecasts. They do not imply that five

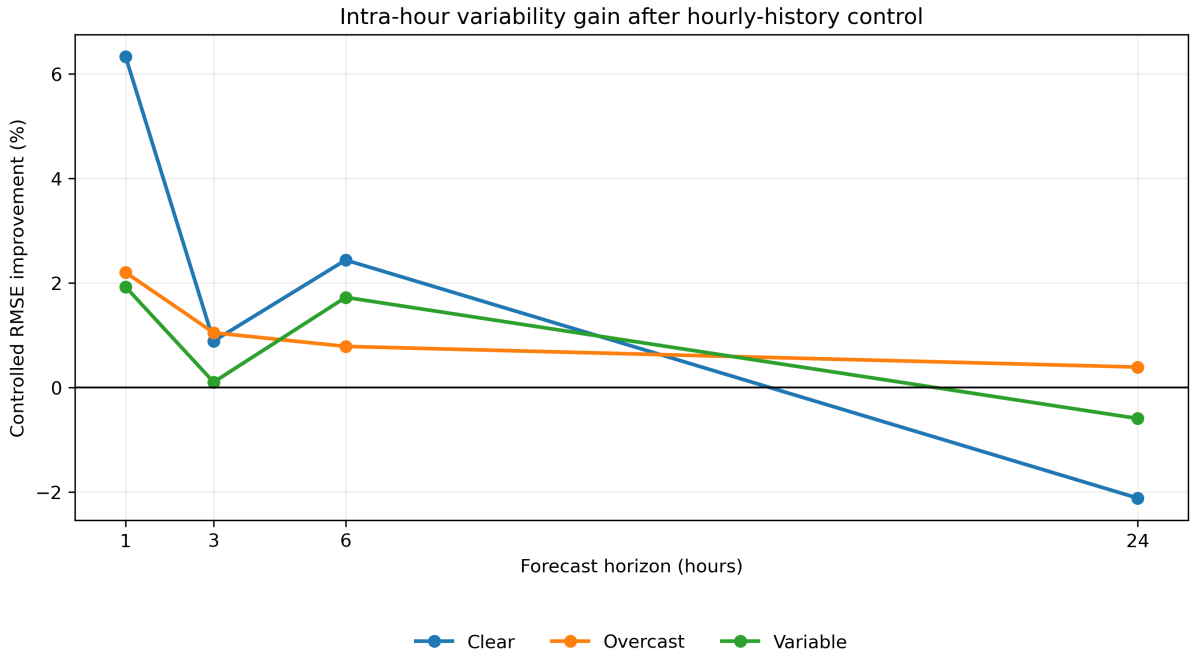


Figure 4.17: Controlled intra-hour gain by daylight weather regime, from the predeclared history-versus-combined comparison.

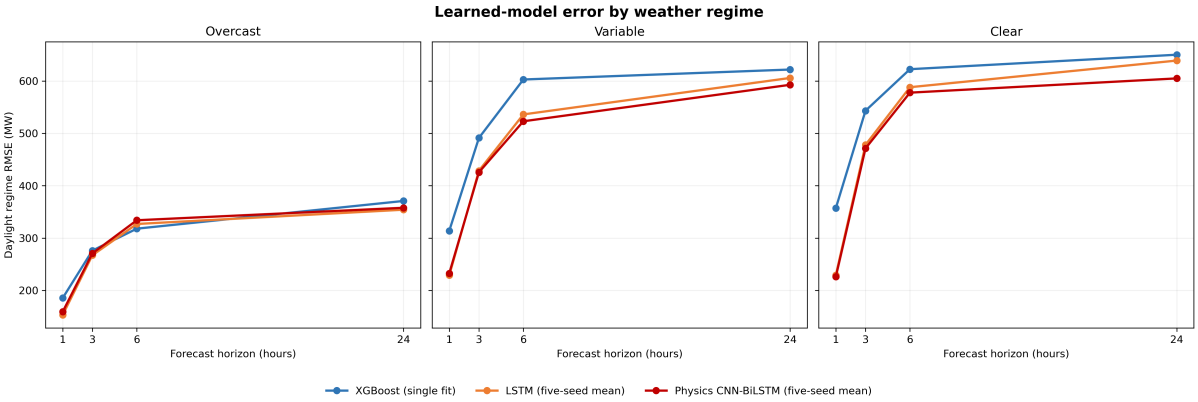


Figure 4.18: Learned-model errors across descriptive daylight weather regimes.

seeds fully characterise the distribution of training outcomes or that averaging will preserve its advantage in another year.

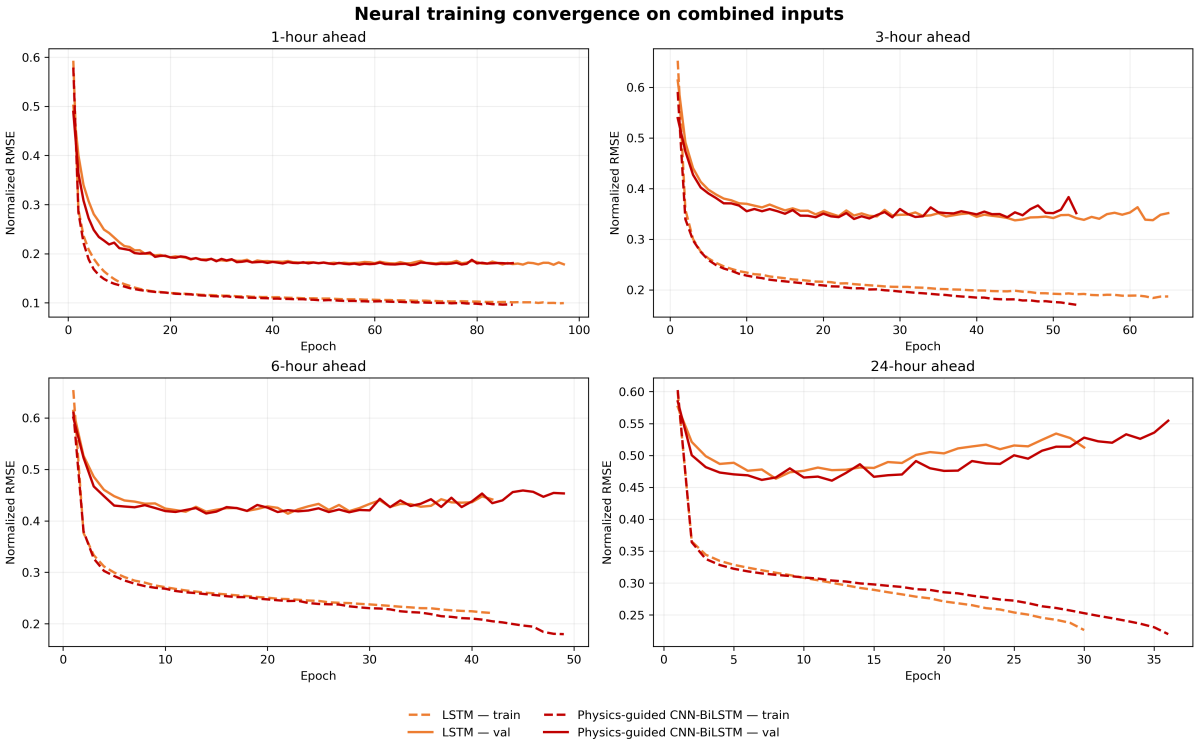


Figure 4.19: Validation convergence of the neural model fits.

The inverse-validation-RMSE ensemble is balanced rather than dominated by one component, yet it never supplies the best combined-input point estimate. Its RMSE is 163.73, 305.72, 375.28 and 407.67 MW. It improves on XGBoost, but the strongest neural model at each horizon remains better. The ensemble is therefore retained as descriptive evidence, not promoted as the final architecture.

Weights remain close to one third because validation errors are similar enough that inverse-RMSE weighting makes only modest adjustments. This limits both the potential benefit and the risk of any one component dominating. The observed result is closer to variance smoothing than specialised regime switching. A more complex stacking model would require its own validation design and would not be justified by the current evidence alone.

The representative trajectories and scatter plots show where aggregate metrics originate. One-hour forecasts closely follow the daily envelope, while longer horizons smooth peaks and underpredict some high-output observations. Dense night-time points around zero improve all-hour scores, whereas daylight scatter widens with horizon. These figures are diagnostic examples and density summaries; they are not additional independent tests.

Taken together, the results form a consistent evidence hierarchy. The largest and least ambiguous effect is the value of ordinary hourly generation history. The next is the advantage of either neural procedure over XGBoost at intraday horizons. The controlled five-minute contribution is smaller but repeatable for XGBoost and physics-guided modelling at 1-6 hours. Neural

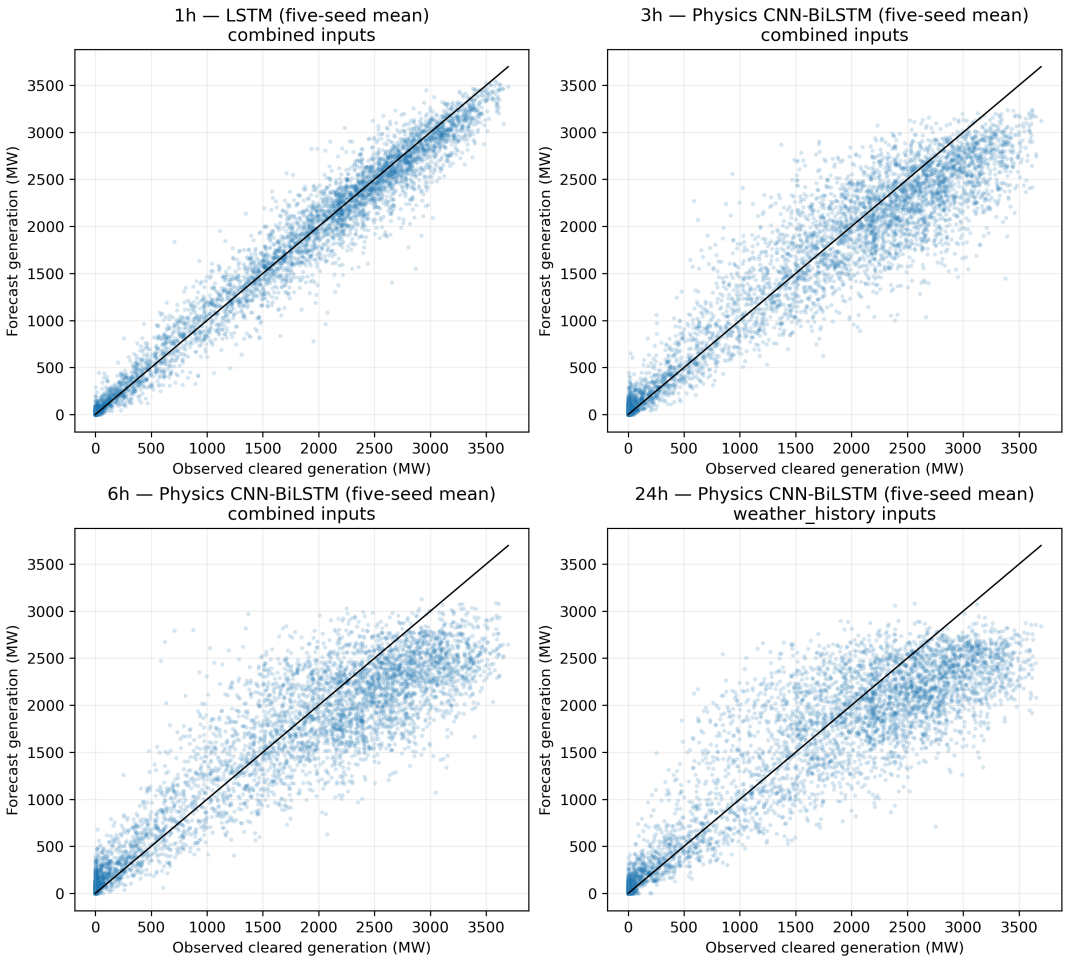


Figure 4.20: Representative observed and predicted values on the 2025 test period. Each panel shows the lowest-RMSE learned forecast for that horizon, so the selected model and input set can differ between panels.

architectural differences are smaller again except at 24 hours. The weighted ensemble does not reverse those rankings.

No single metric changes that ordering. Relative and peak-normalised RMSE agree with absolute RMSE; the output-scale sensitivity check agrees with every controlled direction used in a robust label; and conservative dependence assumptions mainly convert marginal findings to inconclusive rather than producing contradictory positive claims. This convergence supports the stated short-horizon feature conclusion while leaving the day-ahead question open.

The results also show why reporting only the best cell would be misleading. The overall minimum at one hour belongs to LSTM with combined inputs, while the overall 24-hour minimum belongs to physics-guided with hourly history and without the proposed intra-hour variables. A practical forecasting system may therefore use different input sets by horizon, even if it retains one family of models for engineering simplicity.

<!– Citations use the report's global reference numbering. The printed reference list is generated from report/latex/main.bbl. Chapter-local reference lists were removed because they used a superseded numbering and contradicted the printed list. –>

5 Discussion

5.1 Answer to the Research Question

Scale-normalised five-minute intra-hour variability features provide a small, horizon- and model-dependent improvement after ordinary hourly history is controlled. The strongest evidence occurs at 1 hour, where gains are robust for all three learned models and range from 2.52 to 4.88 percent. At 3 and 6 hours, gains remain robust for XGBoost and the physics-guided model but not for LSTM. At 24 hours, no model shows a robust benefit; combined inputs slightly worsen every learned model's point estimate.

This is more precise than saying that five-minute data improve forecasting. Most of the total value of AEMO history comes from the latest hourly level and trailing hourly summaries: for XGBoost these reduce meteorology-only RMSE by 43.21-67.75 percent, whereas controlled intra-hour gains are -0.25 to 4.88 percent. The five-minute summaries are useful incremental signals at short horizons, not substitutes for ordinary generation history.

The model-specific pattern suggests that representation matters. Tree splits can use a variability threshold directly and show robust gains at each intraday horizon. The physics-guided sequence model also benefits consistently at those horizons. LSTM absorbs a robust gain at one hour but not at three or six hours, possibly because its existing sequence representation reconstructs part of the recent dynamics. This explanation is plausible rather than established: hidden-state analysis or a purpose-built interaction experiment would be needed to identify the mechanism.

The 24-hour reversal is also substantively reasonable. Variability in the latest completed hour can be informative about cloud motion over the next few hours, but is less likely to describe conditions one day later. Meanwhile, noise or regime-specific associations learned from that feature can persist in the model. The results therefore favour horizon-specific input policies instead of an assumption that a richer feature vector is universally preferable.

5.2 The Regime Hypothesis Was Not Supported

The regime hypothesis predicted that the intra-hour gain would be largest under variable cloud, on the reasoning that coarse hourly reanalysis would miss rapid change and the dispersion features would then contribute most. The measured ordering does not support that prediction, and in the horizon where the evidence is strongest it points the other way. At one hour, where the controlled gain is robust for all three learned models, the mean gain is 5.58 percent under clear skies, 2.46 percent overcast and 1.84 percent variable. Clear conditions also give the largest mean gain at six hours. Variable conditions give the largest mean gain at no horizon.

This is reported as a failed prediction rather than a confirmed reversal. The regime breakdown is a secondary subgroup analysis covering roughly 1,700 to 2,000 hours per regime with no multiplicity correction, and it reuses a test year that has already been inspected. A rever-

sal established only by subgroup estimates would not meet the standard applied to the primary comparison, and it should not be given more weight here than the result it qualifies.

With that boundary stated, one account fits the observed direction and is worth testing properly. Under broken cloud the hourly mean already contains the cloud-driven variation within that hour, so a dispersion summary computed from the same intervals is largely redundant with the level it accompanies. Under clear skies the hourly level is close to predictable from solar geometry and calendar position, so residual dispersion continues to carry information about the departure from that expectation. On this reading the feature is most useful where hourly history is least informative by itself. This differs from the original reasoning, which treated variable conditions as the case where hourly history would be least adequate.

Two design features limit how far this can be taken. Eight-site equal-weight averaging smooths the spatial structure that produces real cloud transitions, so the variable regime may be less variable in the model inputs than its label suggests. The clearness-index thresholds at 0.3 and 0.7 are conventional rather than derived, and moving them would reassign hours between regimes. A confirmatory test would predeclare the thresholds and the interaction term, and would report uncertainty within each regime.

The practical consequence is narrow. Nothing in the main finding depends on the regime result, because the primary comparison was predeclared on the full sample and is unaffected by how hours are grouped afterwards. What changes is the interpretation offered to a reader: the value of within-hour dispersion is not concentrated where cloud is most active, and the case for collecting five-minute data rests on its short-horizon contribution overall rather than on a claim about cloud regimes.

5.3 Model Interpretation

The neural models generally outperform XGBoost, but their relative ordering depends on horizon. LSTM has the best 1-hour point estimate; the physics-guided CNN-BiLSTM has the best combined-input result at 3, 6 and 24 hours. Pairwise tests find no robust neural difference at 1-6 hours, so choosing between them solely from those small gaps would overstate the evidence. The physics-guided model's robust 24-hour advantage over both XGBoost and LSTM is the clearest architectural result.

That result still belongs to a composite intervention. The model differs from LSTM in convolution, bidirectionality, pooling, head size and two loss penalties. The experiment neither isolates a physics term nor proves a physical mechanism. Its label describes how prior knowledge enters the implemented loss; it is not evidence that the network obeys a complete PV power model.

The ensemble illustrates a similar distinction between diversification and accuracy. Balanced validation weights improve on XGBoost, but do not beat the strongest neural forecast.

No failure-injection, missing-model or cross-year experiment tests operational resilience. On the present objective, the ensemble adds computation without a point-accuracy advantage.

Model choice should consequently account for evidence strength and maintenance cost. At one hour, LSTM's 0.28 MW point advantage over physics-guided is negligible relative to both test uncertainty and seed variation. At three and six hours, physics-guided advantages of 2.37 and 6.11 MW are likewise not robust. A simpler LSTM could be preferred if latency or maintenance dominates. At 24 hours, the physics-guided advantage is larger and robust, but its best input set excludes the intra-hour variables. One global deployment configuration would trade simplicity against small horizon-specific improvements.

5.4 Baselines and Horizon Effects

Persistence is a demanding 24-hour reference because solar output has strong daily periodicity. The physics-guided combined model improves its RMSE by 5.38 percent, but the corrected baseline comparison is not robust at the conservative dependence setting. This is not a contradiction: a positive point estimate can remain uncertain once weekly serial dependence and family-wise multiplicity are respected.

At 1-6 hours, persistence deteriorates as the target moves across the solar ramp. SARIMA is correspondingly strong and provides a more informative short-horizon comparator. Both neural models robustly outperform it at all horizons; XGBoost does not robustly do so at 1 hour. Reporting all three baselines prevents a favourable conclusion from depending on one weak reference.

This baseline structure also changes the meaning of skill. A 77.38 percent improvement over six-hour persistence sounds dramatic, but persistence is particularly unsuitable across a solar-cycle transition, and the same forecast improves less dramatically against SARIMA. Conversely, a small day-ahead persistence gain does not mean the absolute forecast is nearly perfect; it means that copying the latest hour to the same clock time one day later already captures much of the deterministic daily pattern. Absolute MW error and reference-relative skill must therefore be reported together.

5.5 Relation to Previous Research

The result is consistent with work emphasising complementary data sources and recent generation history [13], while narrowing the claim through a controlled ablation. Prior research on intra-hour irradiance variability motivates the feature design [5], but this project forecasts regional market-cleared generation rather than site irradiance. Its effect sizes and mechanism should not be transferred directly.

MERRA-2 supports a reproducible historical benchmark [7], but it is retrospective reanalysis rather than an operational weather forecast available at the original issue time. Consequently, the study evaluates conditional forecasting skill with historical weather fields; it does

not quantify performance under live numerical weather prediction latency or error. Likewise, the physics-guided literature [18] supports adding domain constraints, but the present loss is narrower than an irradiance-to-power physical model. Sequence-to-sequence methods [8] are also not reproduced because each horizon uses an independent scalar-output model.

The benchmark also differs from many plant-level PV studies in spatial and institutional scale. Regional averaging can reduce idiosyncratic plant noise, but NSW-wide clouds, capacity additions and dispatch constraints introduce other forms of non-stationarity. Market-cleared generation is valuable for system-facing forecasting, yet it is not interchangeable with irradiance, AC power at a fixed plant or behind-the-meter demand. Agreement with earlier algorithm rankings should therefore be interpreted as convergence across related tasks, not direct replication.

5.6 Strengths and Limitations

The main strengths are a chronological held-out year, exact elapsed-time alignment, prevention of gap-bridging sequences, conservative duplicate handling, common evaluation windows, five neural seeds and dependence-sensitive inference. The four feature sets distinguish weather, ordinary history and intra-hour dispersion. Output-scale sensitivity and three bandwidths make the main conclusion less dependent on one normalisation or variance assumption. The final notebook also records a clean Apple Silicon MPS execution with package versions.

Several limitations bound the conclusions. First, 2025 is one test year. Moving-block intervals describe uncertainty within that year, not variation across future years, fleet growth or climate regimes. Second, the target excludes most rooftop PV and includes market interventions and curtailment. Third, eight-site mean reanalysis cannot resolve all spatial cloud motion. Fourth, operational feature availability, ingestion delay and live weather forecast quality are untested. Fifth, XGBoost uses one fitted model per block while neural forecasts average five fits, so compute budgets are not equal, and its hyperparameters are tuned on combined inputs and reused for ablations, which improves comparability but may not maximise each feature set. Sixth, clipping predictions at zero affects a non-trivial share of outputs, especially at night. Finally, regime analyses and feature importance are descriptive and do not establish causality.

Inference remains conditional on several choices. Moving blocks approximate serial dependence but cannot reproduce every seasonal or regime transition. Bonferroni correction is conservative and reduces power, especially when tests are correlated, yet it offers a transparent guard against selective reporting. The intersection rule is more demanding again because one unstable diagnostic changes the label to inconclusive. This conservatism explains cases such as the LSTM six-hour gain, where the weekly-block interval is just positive but corrected Diebold-Mariano significance is not stable. The design prioritises claims likely to survive reasonable analytical choices over maximising the number of positive findings.

The input control is stronger than a broad ablation but is not a randomised causal intervention. Adding two columns can change optimisation paths, tree splits and learned regularisation. The reported gain belongs to the complete training procedure with or without those columns. It should not be interpreted as the physical energy value of variability or as the causal effect of a particular cloud process.

The duplicate policy is the least assumptive feasible choice for the legacy archive. Exact duplicates are collapsed; conflicting timestamps are removed because their dispatch scenario is unknowable. This removes every duplicated timestamp, but it cannot recover the missing normal-dispatch observation and may introduce minor selection effects through the 120 excluded hourly bins.

Alternative duplicate strategies have less defensible consequences. Keeping the first or last row would make the target depend on file ordering, while averaging conflicting rows could create a value corresponding to neither dispatch scenario. Dropping an entire conflicting group loses data but keeps the retained target observations interpretable. Because no excluded hour occurs in the test year, the direct test metrics are not changed by a test-period duplicate choice; the training sample and fitted parameters can still be affected.

5.7 Implications

For this NSW1 task, recent hourly generation should be a core operational input. The two intra-hour summaries are worthwhile at short horizons when five-minute data are reliably available, although their modest incremental effect must be weighed against ingestion complexity. At 24 hours, they should not be assumed beneficial.

The physics-guided model is the strongest single candidate when one architecture must serve all four horizons; LSTM is equally defensible at 1-6 hours on the present statistical evidence and is simpler. Deployment would still require genuine issue-time weather, latency monitoring, missing-data fallback, recalibration under capacity growth and rolling out-of-year validation. RMSE improvements cannot be converted to trading value without prices, participant exposure and an explicit decision model.

A staged deployment study could begin with shadow forecasts. Predictions would be timestamped before target delivery, and every feature would be accompanied by its source publication time, which is the only way to test whether the benchmark survives real issue times. Performance would be monitored by horizon, daylight condition, output level and missing-data state.

Future evaluation should also examine probabilistic and decision-focused outcomes. Calibrated prediction intervals can express weather and model uncertainty that point RMSE cannot. Ramp-event recall, interval coverage and asymmetric cost-weighted loss may matter more than average squared error for some operators. Those objectives require predeclared definitions and stakeholder-specific costs; they cannot be inferred from the present point forecasts.

Interpretability work should focus on stability rather than a single feature-importance chart. XGBoost importance can identify variables used frequently or effectively by one fitted tree ensemble, but correlated lags can exchange importance without changing predictions. Permutation tests grouped by feature family, accumulated local effects and temporal stability checks would better assess whether the intra-hour signal is used consistently. Neural attribution would require analogous perturbation tests evaluated on held-out sequences. Such analyses could help diagnose mechanism, but they would remain observational unless supported by a design that separates weather dynamics from measurement artefacts.

Capacity growth deserves a dedicated treatment as well. Past-only output scaling reduces one form of leakage and makes relative error more comparable through time, yet the model still learns from a fleet whose composition changes. Plant connections, outages and constraint patterns may alter the mapping between regional weather and cleared generation. Incorporating an issue-time capacity series, where reliable, could separate level growth from short-term weather response. Any such variable would need the same availability audit applied to generation history.

Finally, computational reproducibility should be distinguished from numerical identity. The successful Apple Silicon MPS run demonstrates that the self-contained notebook is portable beyond the AMD/ROCm development workstation without guaranteeing bitwise equality across devices. Reproducibility rests instead on recorded versions, fixed seeds, clean execution ordering and agreement of the substantive conclusions under repeated neural fits.

The retained evidence tables further improve auditability by separating point metrics, seed variability, feature contrasts and inferential decisions, so a reader can trace each headline statement to a model-horizon row. This matters most for inconclusive findings, which are easy to lose when only the lowest RMSE is highlighted.

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6 Conclusion

6.1 Summary

This study asked whether scale-normalised five-minute intra-hour variability improves NSW1 solar forecasting after ordinary hourly generation history is controlled. The answer is qualified but positive at short horizons. All three learned models improve robustly at 1 hour. XGBoost and the physics-guided CNN-BiLSTM also improve robustly at 3 and 6 hours. LSTM's intermediate-horizon gains are inconclusive, and no robust benefit is found at 24 hours. Controlled gains are modest, from 1.12 to 4.88 percent where robust, while ordinary hourly history supplies the dominant reduction in error.

On combined inputs, LSTM records the lowest 1-hour RMSE at 155.25 MW. The physics-guided model records the lowest values at 3, 6 and 24 hours: 300.28, 369.17 and 400.81 MW. The two neural models are not robustly different at 1-6 hours; physics-guided is robustly better at 24 hours. It also gives the lowest overall 24-hour feature-set result, 391.49 MW with weather_history, showing that the intra-hour addition is not automatically helpful at day-ahead range.

The ensemble improves on XGBoost but not on the best neural forecast. Baseline comparisons show robust learned-model advantages at 1-6 hours, while persistence remains difficult to beat conclusively at 24 hours. These findings are about 2025 market-cleared utility-scale generation under retrospective reanalysis inputs, not rooftop PV, unconstrained physical production or economic value.

The conclusion also distinguishes point ranking from evidential ranking. Although one model has the lowest RMSE in each block, the paired tests show that LSTM and physics-guided performance is effectively unresolved at 1-6 hours under the stated decision rule. The report therefore does not claim that the more complex neural architecture is universally superior. Its robust advantage appears at 24 hours, where its weather_history input set is the overall best configuration.

6.2 Contributions

The project contributes a self-contained regional benchmark joining AEMO five-minute generation with hourly weather, a duplicate and alignment policy that prevents ambiguous timestamps and gap-spanning sequences, and a controlled feature design that separates ordinary hourly history from scale-normalised intra-hour variability. It evaluates three learned families and three baselines on common windows, averages five neural seeds, and requires agreement across block-bootstrap intervals, multiple Diebold-Mariano bandwidths and an alternative output scale before calling a feature result robust.

The final report and notebook are also reproducible artefacts. The saved notebook completed without errors from a fresh kernel on Apple Silicon using MPS and records the exact environment and execution time.

Methodologically, the work demonstrates why forecast evaluation should preserve clock semantics and dependence. Exact elapsed-time joins stop missing observations from silently redefining a horizon. Contiguous sequences stop a recurrent model from treating gaps as adjacent hours. Multiple dependence bandwidths reveal when a nominally significant result relies on a short-memory assumption. These controls are transferable to other energy time-series studies even where the particular model rankings differ.

Two analytical findings extend beyond the headline answer. First, the factorial design separates the frequency of a data source from its incremental information. The five-minute columns contribute between 0.95 and 4.41 percent against a meteorology-only baseline but only 4.88 percent at one hour once hourly history is present, and the two estimates converge because a completed hour's mean already carries much of the variation inside it. Reporting either figure alone would misrepresent the same feature set, which is an argument for controlled ablations over single-baseline comparisons in data-integration studies generally.

Second, the training records show that the neural advantage narrows with horizon for a reason visible in the fits themselves. Validation error bottomed out after roughly sixty epochs at one hour but within about nine to twelve epochs at day-ahead range, so the recurrent models extracted far less structure from an identical input window as the horizon lengthened. This reframes a model-ranking result as a training-behaviour result and suggests that horizon-specific stopping rules deserve the same attention as horizon-specific inputs.

6.3 Recommendations and Future Work

An operational extension should first replace reanalysis with archived issue-time weather forecasts and reproduce the comparison across multiple held-out years and other NEM regions. A deployment pipeline should define source latency, late and conflicting record handling, missing-feature fallbacks, capacity-change monitoring and retraining triggers.

Further experiments should isolate the physics-guided model's components with matched CNN-BiLSTM controls, equalise tuning and compute budgets, and test whether richer intra-hour representations add value beyond the two summary statistics. Probabilistic forecasts should be assessed with calibration, coverage and sharpness rather than point RMSE alone. Regime-specific and ramp-event evaluation should be predeclared and powered for subgroup inference.

Multi-year rolling-origin evaluation is the most important next validation step. It would show whether the controlled feature effect persists through different weather years and stages of fleet growth, and it would allow uncertainty across years rather than only within 2025. Cross-region evaluation could then separate NSW-specific conclusions from more general properties of utility-scale solar forecasting.

Operational research should preserve the same conservative data rules. Conflicting records need an auditable quarantine path; identical re-deliveries can be de-duplicated deterministically;

and incomplete hours should trigger a documented fallback rather than silent zero filling. Actual publication timestamps should replace assumptions about immediate availability. These changes would convert the retrospective benchmark into a test of feasible forecast issuance.

Finally, forecast value should be measured in the decision context. An operator may prioritise ramp warnings and reliability, while a participant may face asymmetric imbalance costs. The modest controlled RMSE gains reported here could be valuable, negligible or even counterproductive under those objectives. Connecting predictions to decisions requires price, constraint and exposure data beyond the present scope.

The recommended sequence is therefore evidence-led. First, preserve hourly generation history and the strong baselines. Second, add the two scale-normalised intra-hour summaries for intraday horizons and validate them continuously. Third, retain a horizon-specific option to omit them at 24 hours. Fourth, compare LSTM and physics-guided implementations under measured latency and maintenance constraints rather than treating small RMSE gaps as decisive. Only after these steps should a more complex ensemble or richer high-frequency representation be introduced.

This sequence also protects the main scientific conclusion from changes in deployment architecture. Even if a future transformer, probabilistic model or improved weather source lowers absolute error, the controlled question can be repeated: does five-minute variability add information beyond the latest hourly level and trailing history? Maintaining that control will make future results comparable with the present benchmark.

The study does not resolve every question raised by high-frequency solar data. It does establish a defensible baseline answer using the information available: frequency alone is not the contribution. The useful contribution is the additional, forecast-time-available signal that remains after lower-frequency history has been held constant, evaluated under temporal dependence and multiplicity. That framing is likely to remain valuable as datasets and architectures change.

6.4 Closing Statement

Recent hourly generation is the central predictive input in this benchmark. Five-minute variability adds smaller but repeatable short-horizon information, while its day-ahead value is unsupported. The result favours controlled feature integration over claims that higher-frequency data or greater model complexity must improve every horizon.

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Appendix A: Data Sources and Licences

The generation source is AEMO DISPATCHREGIONSUM for NSW1, retrieved through NEMOSIS. The legacy export lacks intervention labels. Exact-value duplicate timestamps are collapsed; conflicting timestamp groups are dropped because the normal-dispatch observation cannot be identified. Hourly meteorology comes from renewables.ninja for Sydney, Newcastle, Wollongong, Central Coast, Tamworth, Bathurst, Orange and Dubbo. Source attribution and access conditions should accompany redistribution of archived data. The simulated renewables.ninja PV series is excluded from prediction inputs.

Appendix B: Reproducibility Configuration

Table B.1 records the final training and evaluation configuration.

Table B.1: Reproducibility configuration summary.

Component	Setting
Chronological split	train September 2020 to 2023; validation 2024; test 2025
Feature sets	weather (11), weather_intra (13), weather_history (14), combined (16)
Forecast horizons	1, 3, 6 and 24 hours; direct scalar model per horizon
Sequence construction	contiguous 24-hour windows; exact elapsed-time target matching
LSTM	one layer; 64 hidden units
Physics-guided model	Conv1D 32 filters, kernel 3; max pool; bidirectional LSTM 64; head 128
Physics-guided loss	MSE plus 0.1 times negative-output and night-time zero penalties
Optimisation	batch 512; learning rate 0.001; maximum 100 epochs
Early stopping	patience 20; best validation checkpoint restored
Random seeds	42, 123, 456, 789 and 1024
XGBoost	30 Optuna trials per horizon on combined inputs; settings reused across sets
Ensemble	inverse validation-RMSE weights across three learned models
Significance testing	Diebold-Mariano bandwidths h-1, 24 and 168; family-wise Bonferroni
Bootstrap CIs	95%; 2,000 moving-block resamples; 24, 48 and 168-hour blocks
Final hardware	Apple Silicon arm64; PyTorch MPS
Final environment	conda spunky; Python 3.12.12; PyTorch 2.5.1; XGBoost 3.0.1
Execution	13 September 2026; clean kernel; 30 code cells; no errors or stderr
Notebook SHA-256	2176aca9641fe9b65eb6bd881d9b14fbaa3887c8f2b7e8d2940d64e34285544a